

Chapter 18

Regulation of Gene Expression

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

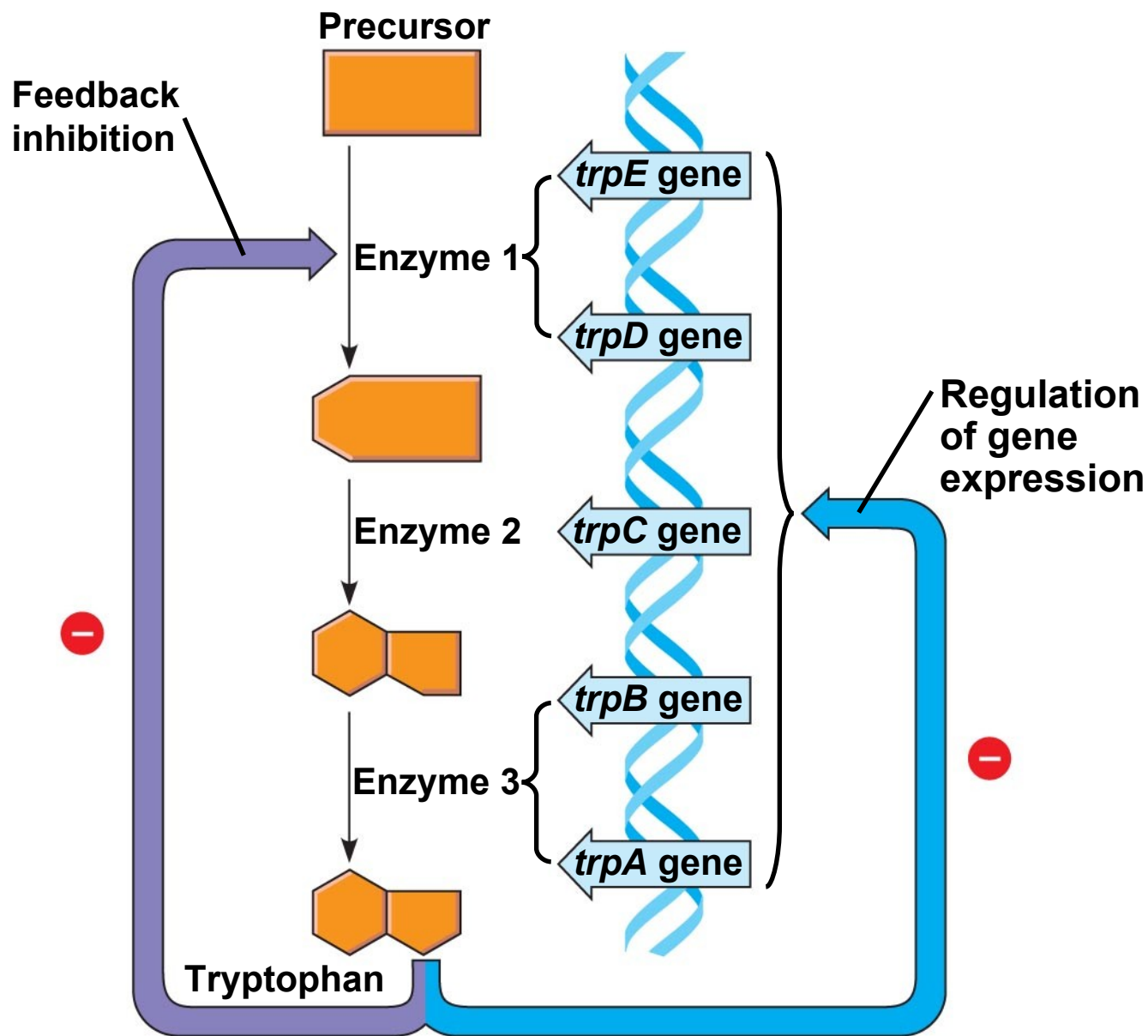
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Overview: Conducting the Genetic Orchestra

- Prokaryotes and eukaryotes alter gene expression in response to their changing environment
- In multicellular eukaryotes, gene expression regulates development and is responsible for differences in cell types
- RNA molecules play many roles in regulating gene expression in eukaryotes

Concept 18.1: Bacteria often respond to environmental change by regulating transcription

- Natural selection has favored bacteria that produce only the products needed by that cell
- A cell can regulate the production of enzymes by feedback inhibition or by gene regulation
- Gene expression in bacteria is controlled by the operon model



(a) Regulation of enzyme activity

(b) Regulation of enzyme production

Operons: The Basic Concept

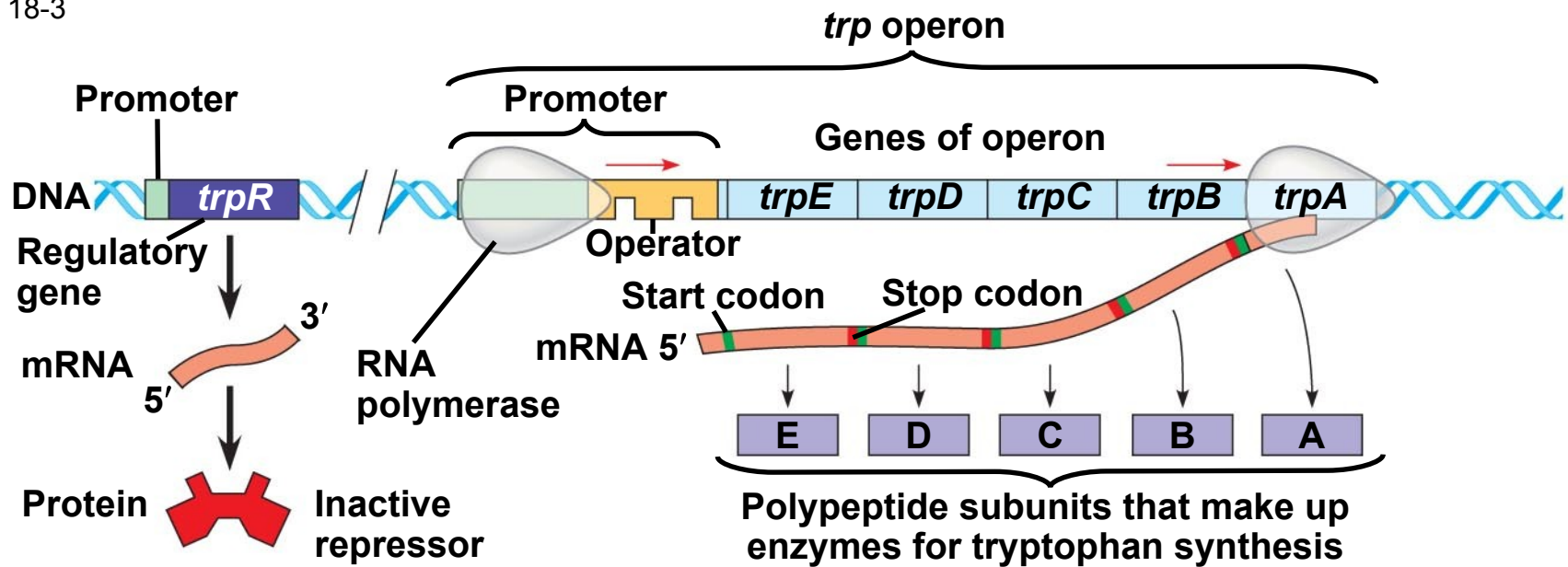
- A cluster of functionally related genes can be under coordinated control by a single on-off “switch”
- The regulatory “switch” is a segment of DNA called an **operator** usually positioned within the promoter
- An **operon** is the entire stretch of DNA that includes the operator, the promoter, and the genes that they control

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- The operon can be switched off by a protein **repressor**
 - The repressor prevents gene transcription by binding to the operator and blocking RNA polymerase
 - The repressor is the product of a separate **regulatory gene**

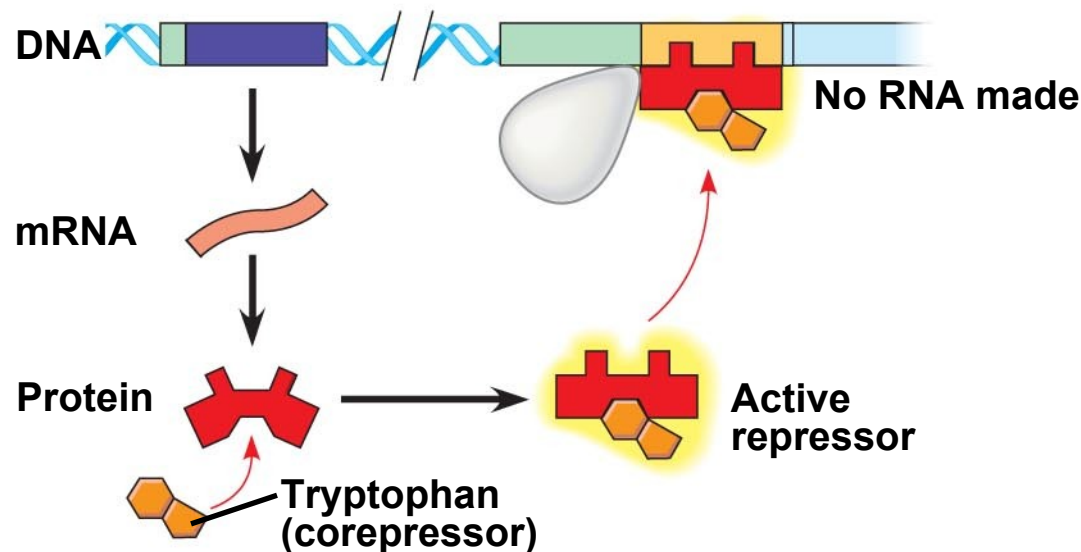
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- The repressor can be in an active or inactive form, depending on the presence of other molecules
 - A **corepressor** is a molecule that cooperates with a repressor protein to switch an operon off
 - For example, *E. coli* can synthesize the amino acid tryptophan

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- By default the *trp* operon is on and the genes for tryptophan synthesis are transcribed
 - When tryptophan is present, it binds to the *trp* repressor protein, which turns the operon off
 - The repressor is active only in the presence of its corepressor tryptophan; thus the *trp* operon is turned off (repressed) if tryptophan levels are high

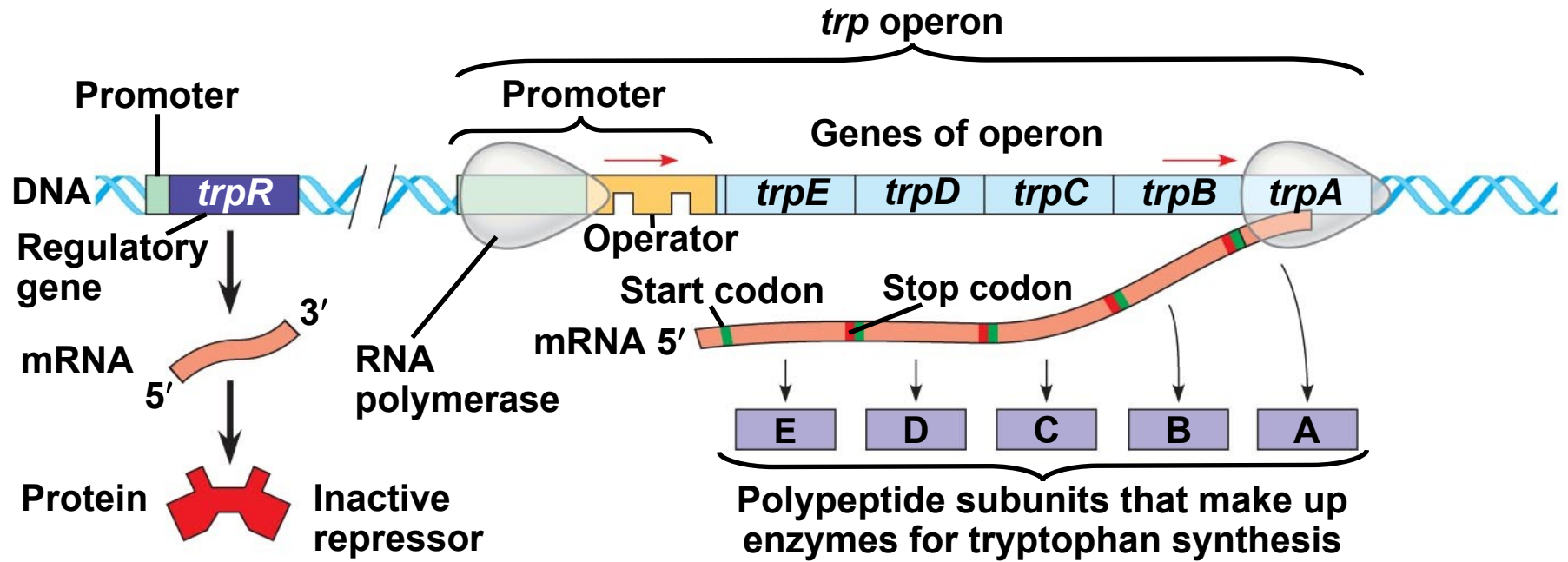
Fig. 18-3



(a) Tryptophan absent, repressor inactive, operon on

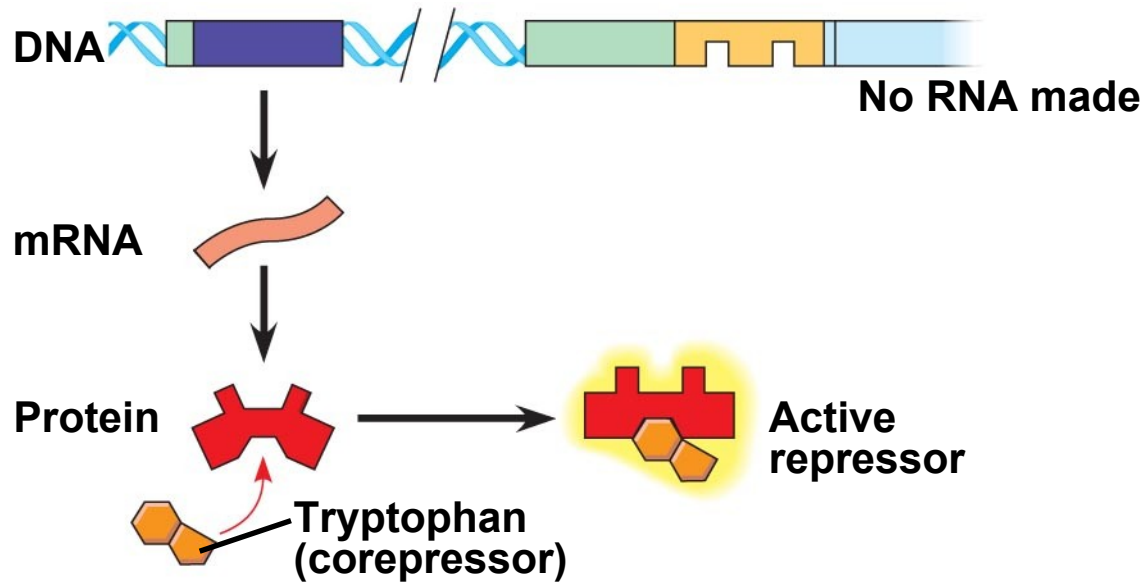


(b) Tryptophan present, repressor active, operon off



(a) Tryptophan absent, repressor inactive, operon on

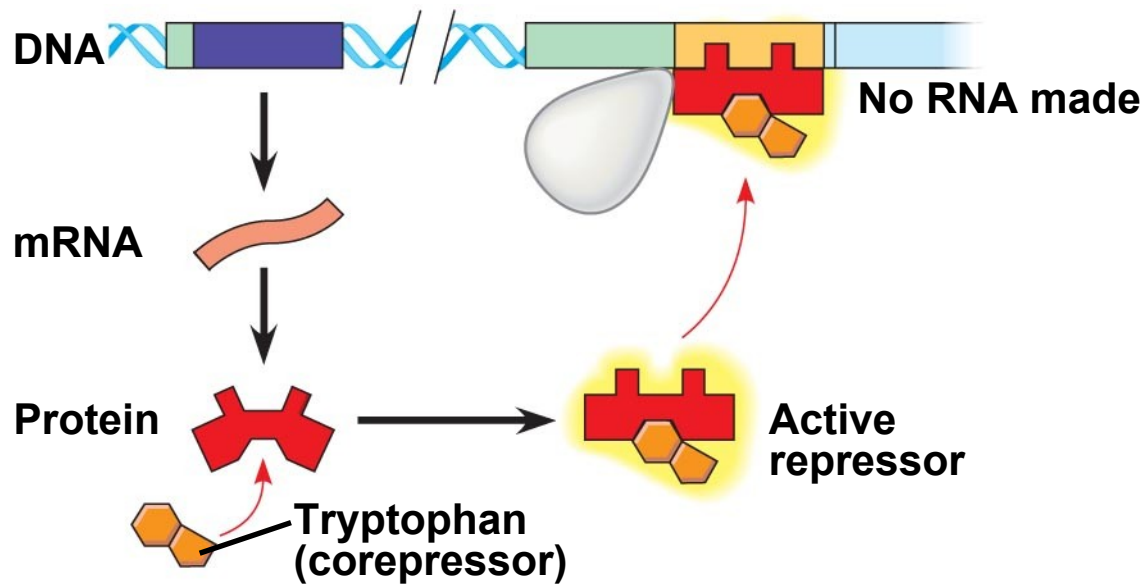
Fig. 18-3b-1



(b) Tryptophan present, repressor active, operon off

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Fig. 18-3b-2



(b) Tryptophan present, repressor active, operon off

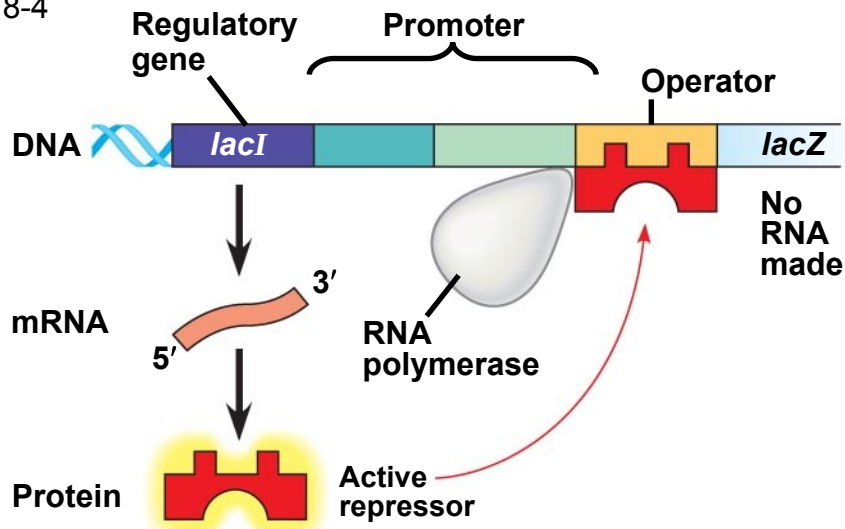
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Repressible and Inducible Operons: Two Types of Negative Gene Regulation

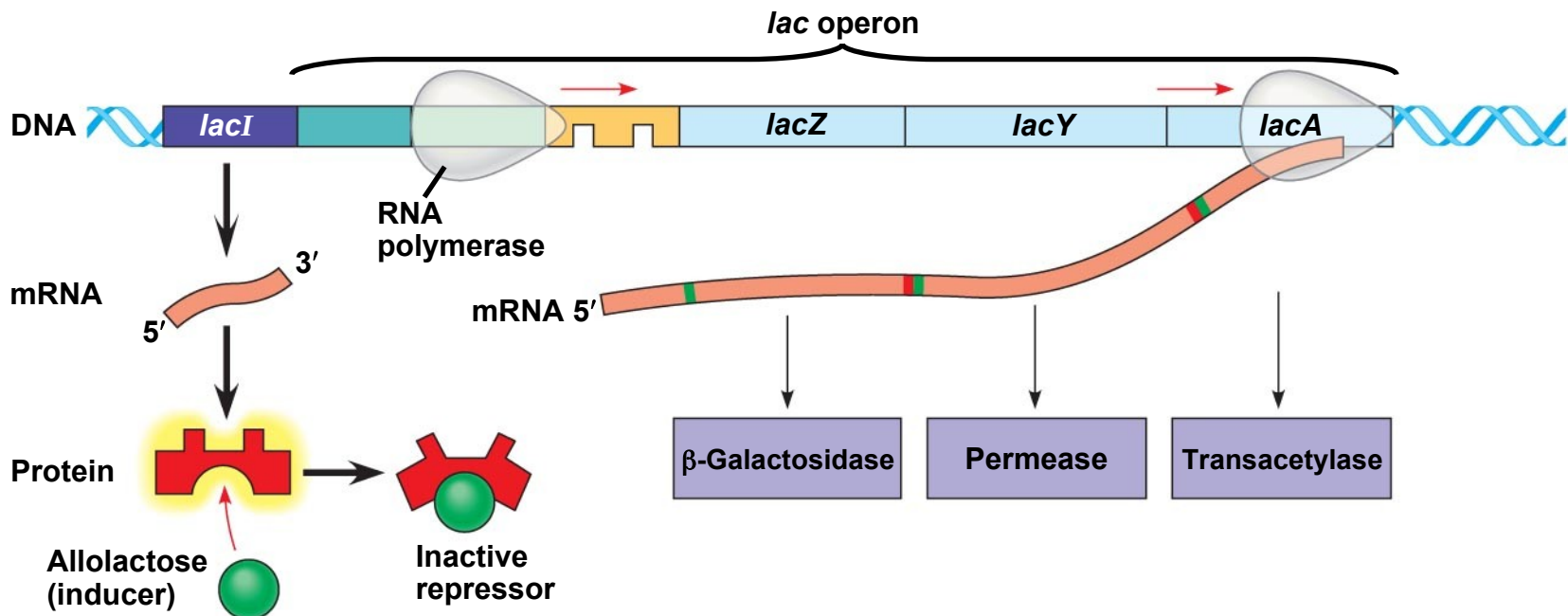
- A repressible operon is one that is usually on; binding of a repressor to the operator shuts off transcription
- The *trp* operon is a repressible operon
- An inducible operon is one that is usually off; a molecule called an inducer inactivates the repressor and turns on transcription

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- The *lac* operon is an inducible operon and contains genes that code for enzymes used in the hydrolysis and metabolism of lactose
 - By itself, the *lac* repressor is active and switches the *lac* operon off
 - A molecule called an **inducer** inactivates the repressor to turn the *lac* operon on

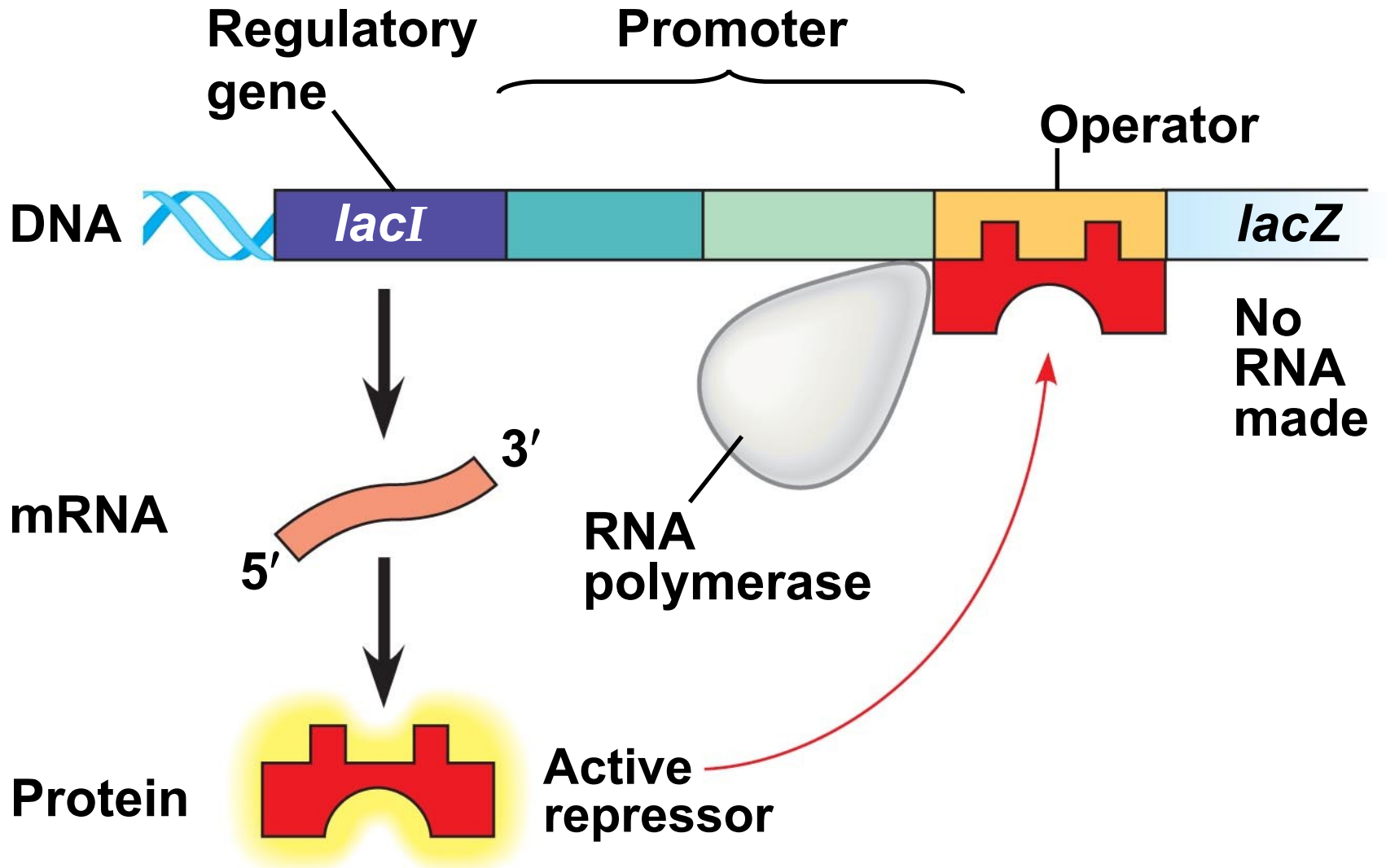
Fig. 18-4



(a) Lactose absent, repressor active, operon off

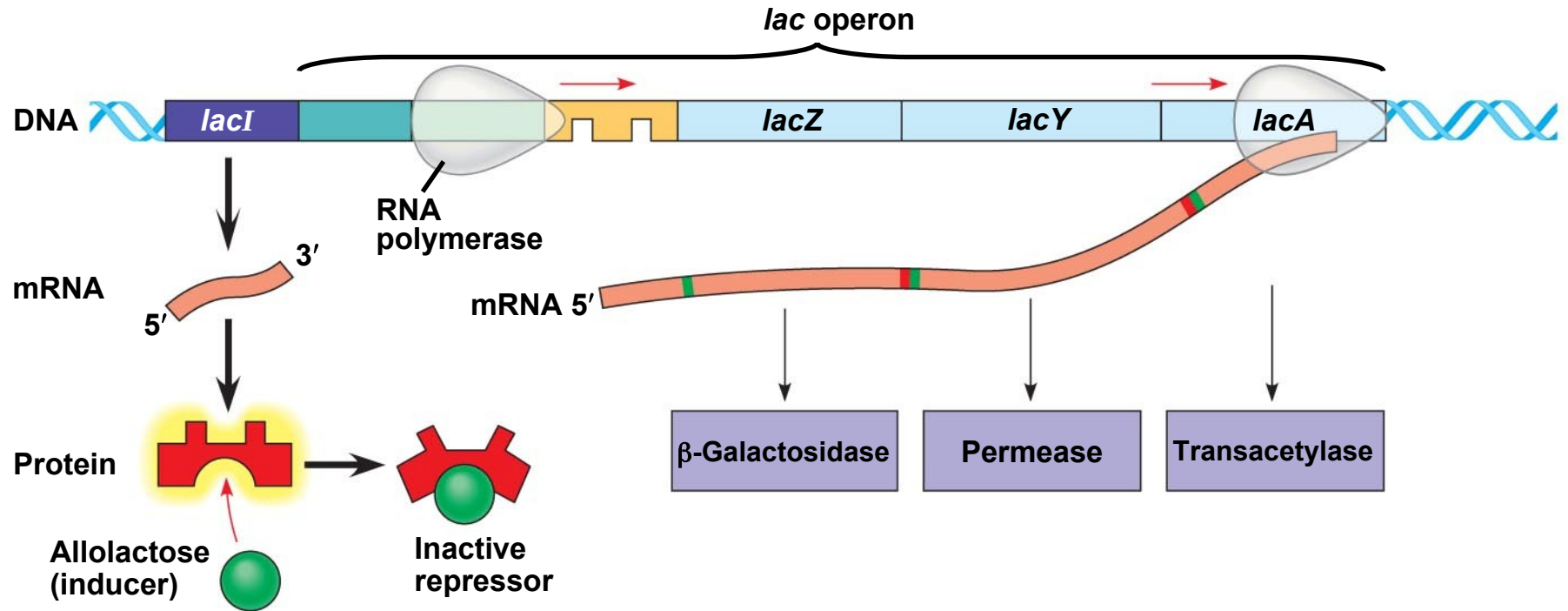


(b) Lactose present, repressor inactive, operon on



(a) Lactose absent, repressor active, operon off

Fig. 18-4b



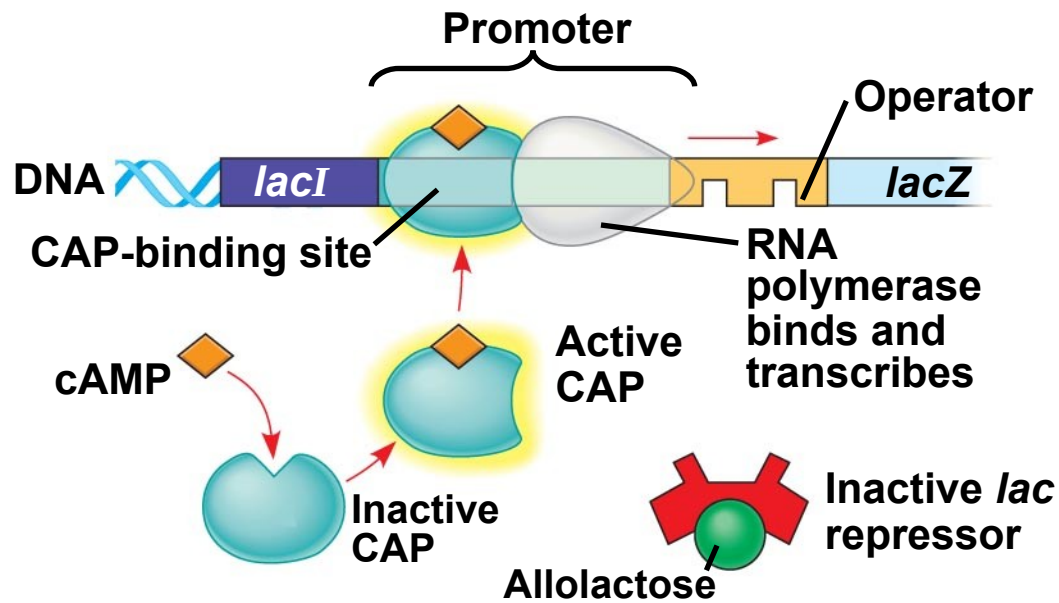
(b) Lactose present, repressor inactive, operon on

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- Inducible enzymes usually function in catabolic pathways; their synthesis is induced by a chemical signal
 - Repressible enzymes usually function in anabolic pathways; their synthesis is repressed by high levels of the end product
 - Regulation of the *trp* and *lac* operons involves negative control of genes because operons are switched off by the active form of the repressor

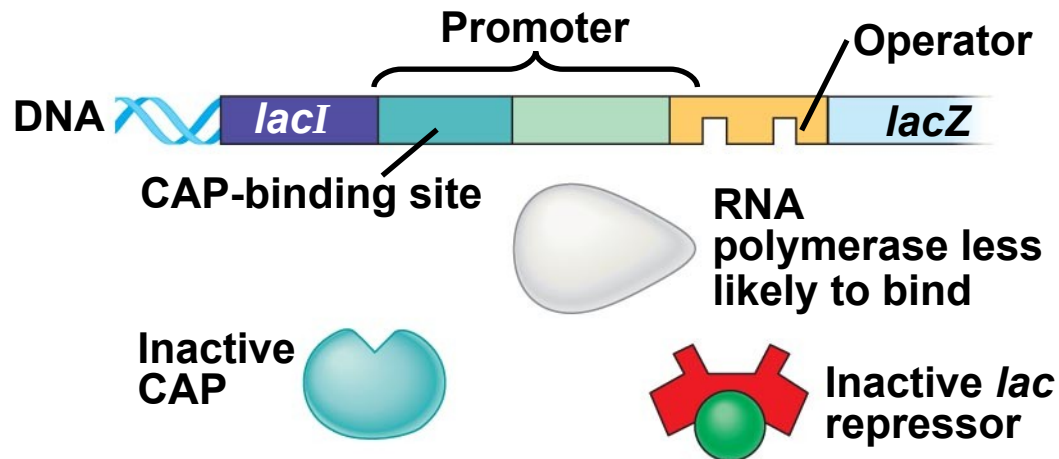
Positive Gene Regulation

- Some operons are also subject to positive control through a stimulatory protein, such as catabolite activator protein (CAP), an **activator** of transcription
- When glucose (a preferred food source of *E. coli*) is scarce, CAP is activated by binding with **cyclic AMP**
- Activated CAP attaches to the promoter of the *lac* operon and increases the affinity of RNA polymerase, thus accelerating transcription

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- When glucose levels increase, CAP detaches from the *lac* operon, and transcription returns to a normal rate
 - CAP helps regulate other operons that encode enzymes used in catabolic pathways



(a) Lactose present, glucose scarce (cAMP level high): abundant *lac* mRNA synthesized



(b) Lactose present, glucose present (cAMP level low): little *lac* mRNA synthesized

Concept 18.2: Eukaryotic gene expression can be regulated at any stage

- All organisms must regulate which genes are expressed at any given time
- In multicellular organisms gene expression is essential for cell specialization

Differential Gene Expression

- Almost all the cells in an organism are genetically identical
- Differences between cell types result from **differential gene expression**, the expression of different genes by cells with the same genome
- Errors in gene expression can lead to diseases including cancer
- Gene expression is regulated at many stages

Fig. 18-6

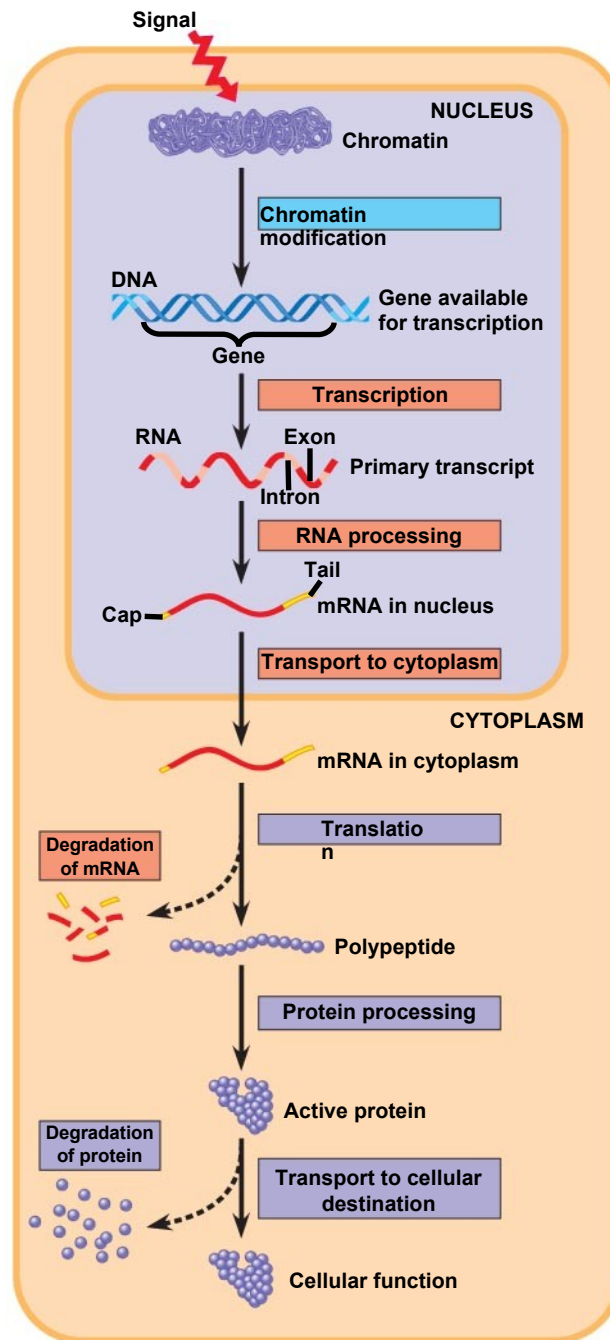


Fig. 18-6a

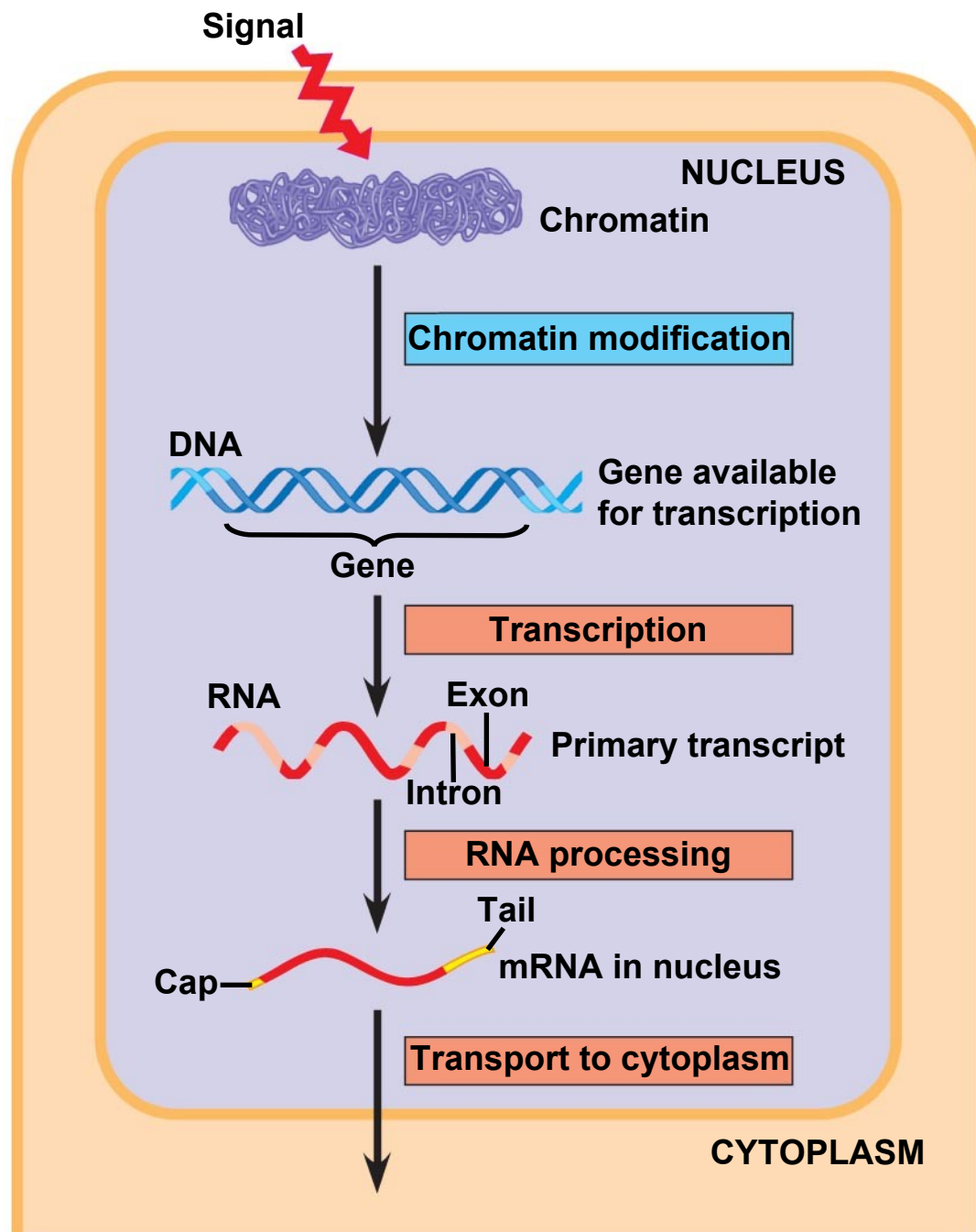
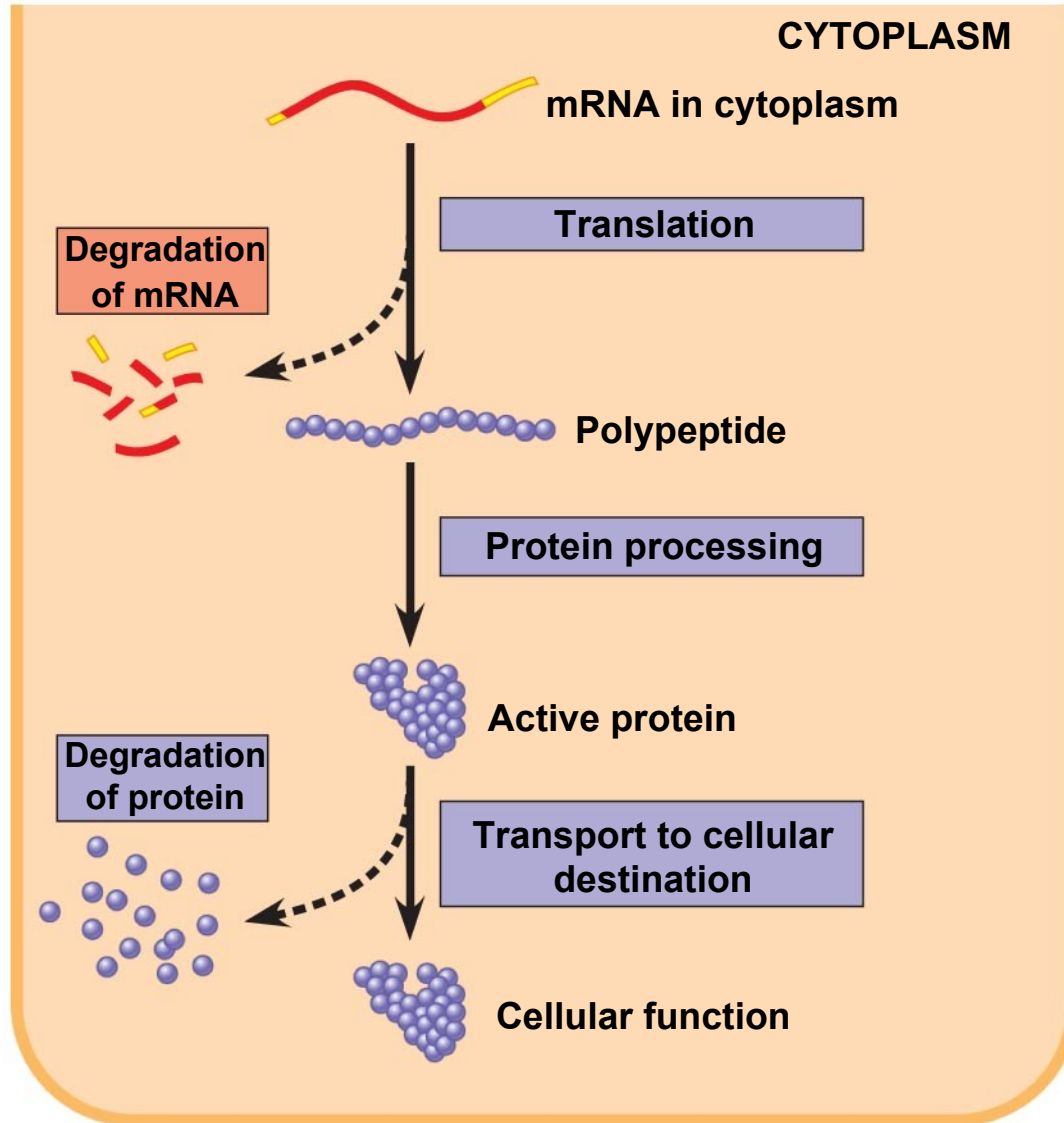


Fig. 18-6b



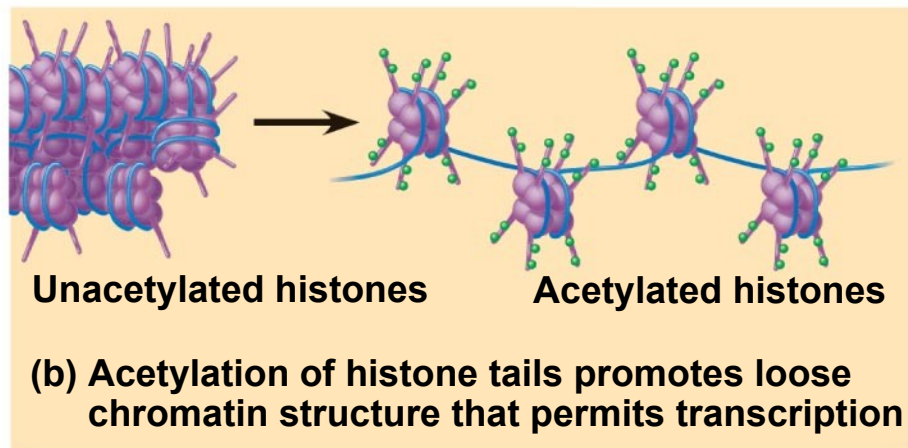
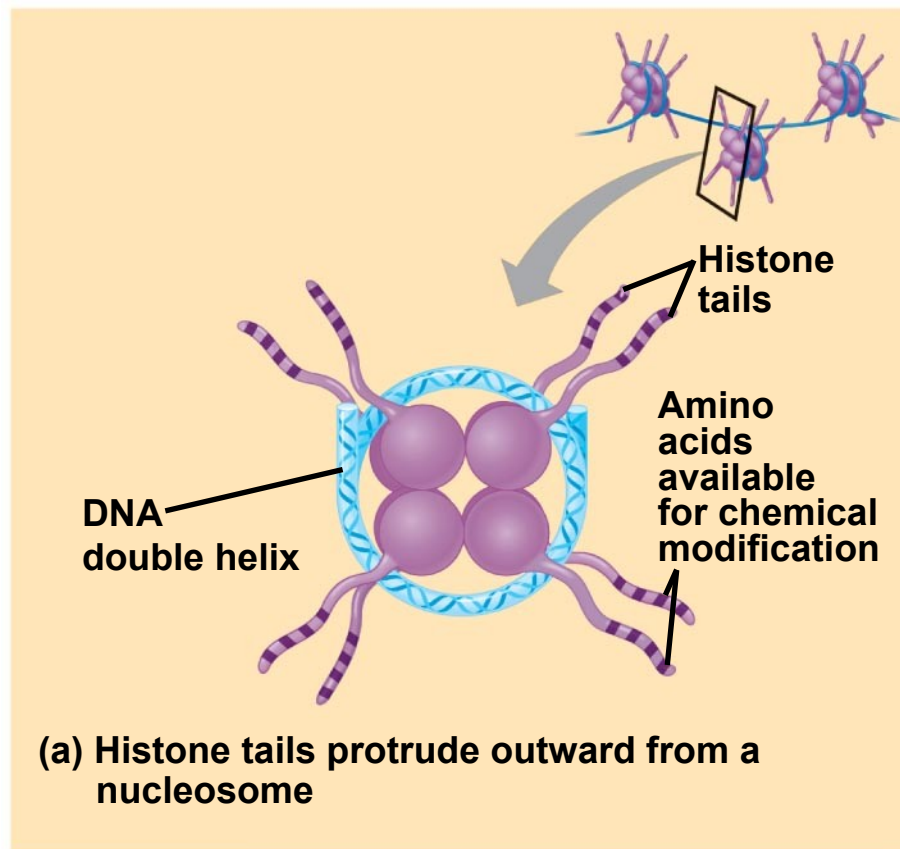
Regulation of Chromatin Structure

- Genes within highly packed heterochromatin are usually not expressed
- Chemical modifications to histones and DNA of chromatin influence both chromatin structure and gene expression

Histone Modifications

- In **histone acetylation**, acetyl groups are attached to positively charged lysines in histone tails
- This process loosens chromatin structure, thereby promoting the initiation of transcription
- The addition of methyl groups (methylation) can condense chromatin; the addition of phosphate groups (phosphorylation) next to a methylated amino acid can loosen chromatin

Fig. 18-7



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- The histone code hypothesis proposes that specific combinations of modifications help determine chromatin configuration and influence transcription

DNA Methylation

- DNA methylation, the addition of methyl groups to certain bases in DNA, is associated with reduced transcription in some species
- DNA methylation can cause long-term inactivation of genes in cellular differentiation
- In **genomic imprinting**, methylation regulates expression of either the maternal or paternal alleles of certain genes at the start of development

Epigenetic Inheritance

- Although the chromatin modifications just discussed do not alter DNA sequence, they may be passed to future generations of cells
- The inheritance of traits transmitted by mechanisms not directly involving the nucleotide sequence is called **epigenetic inheritance**

Regulation of Transcription Initiation

- Chromatin-modifying enzymes provide initial control of gene expression by making a region of DNA either more or less able to bind the transcription machinery

Organization of a Typical Eukaryotic Gene

- Associated with most eukaryotic genes are **control elements**, segments of noncoding DNA that help regulate transcription by binding certain proteins
- Control elements and the proteins they bind are critical to the precise regulation of gene expression in different cell types

Fig. 18-8-1

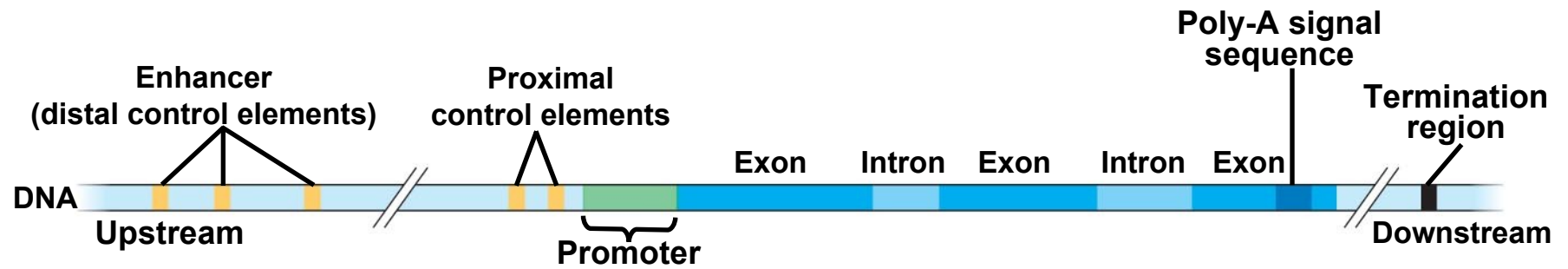


Fig. 18-8-2

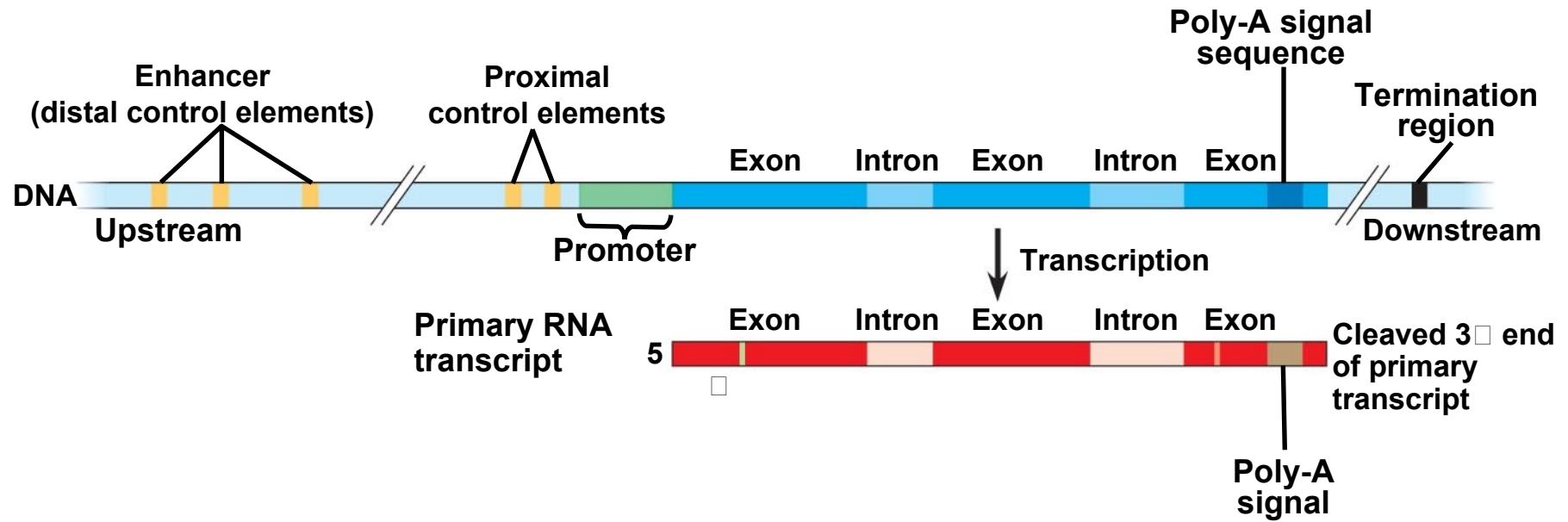
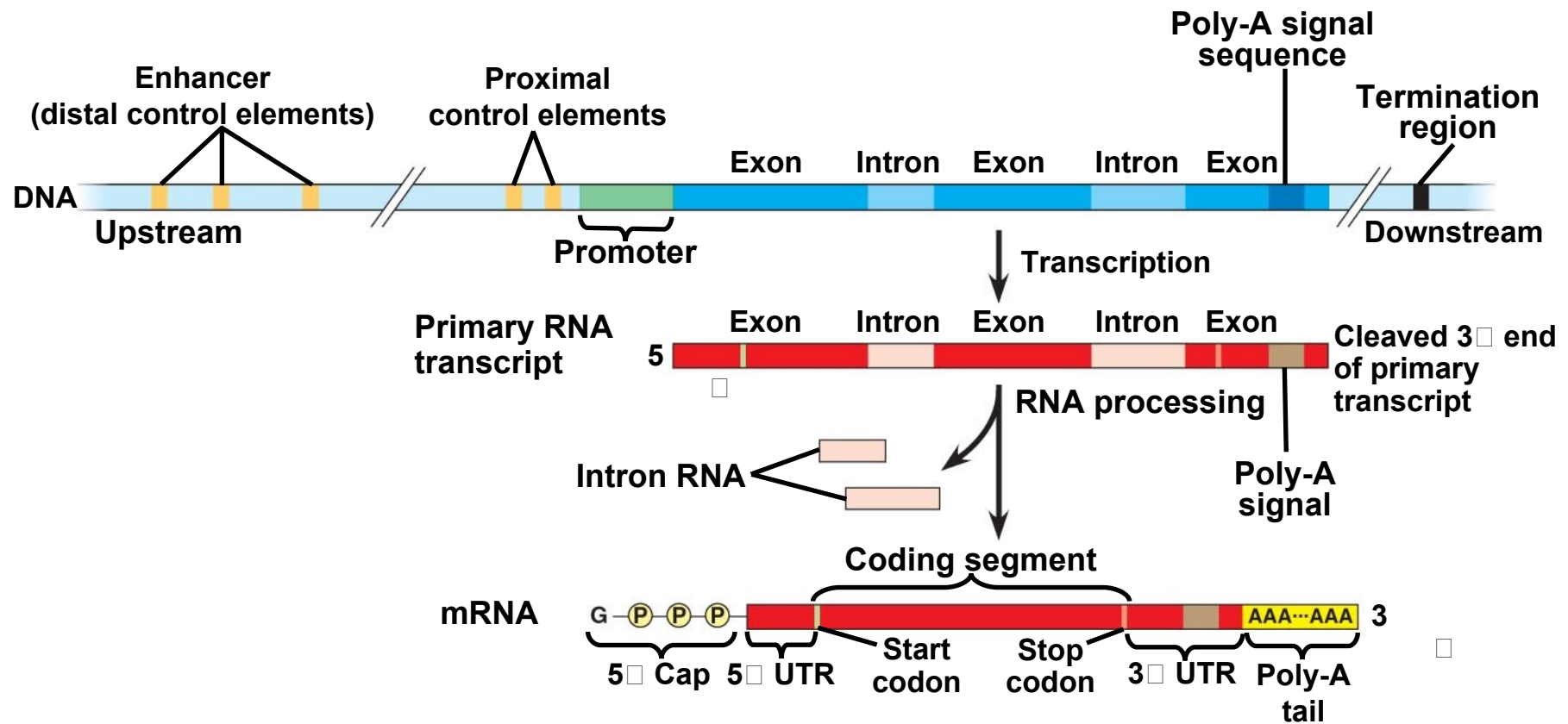


Fig. 18-8-3



The Roles of Transcription Factors

- To initiate transcription, eukaryotic RNA polymerase requires the assistance of proteins called transcription factors
- General transcription factors are essential for the transcription of all protein-coding genes
- In eukaryotes, high levels of transcription of particular genes depend on control elements interacting with specific transcription factors

Enhancers and Specific Transcription Factors

- Proximal control elements are located close to the promoter
- Distal control elements, groups of which are called **enhancers**, may be far away from a gene or even located in an intron

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- An activator is a protein that binds to an enhancer and stimulates transcription of a gene
 - Bound activators cause mediator proteins to interact with proteins at the promoter

Fig. 18-9-1

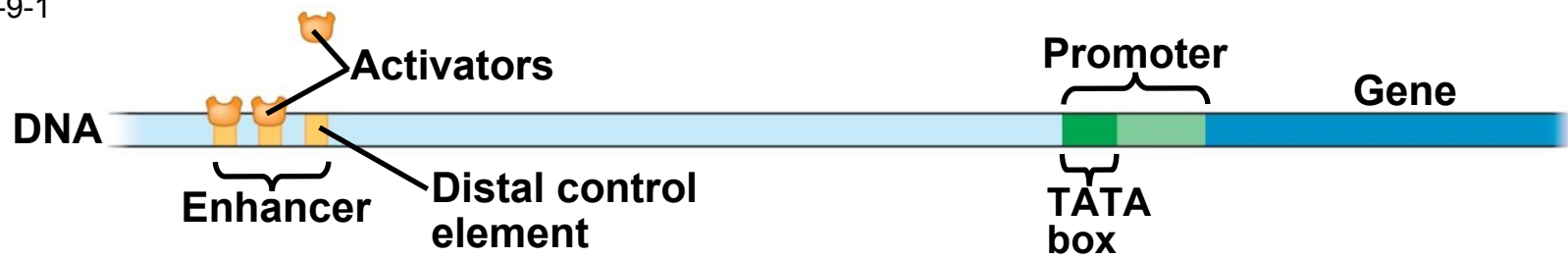


Fig. 18-9-2

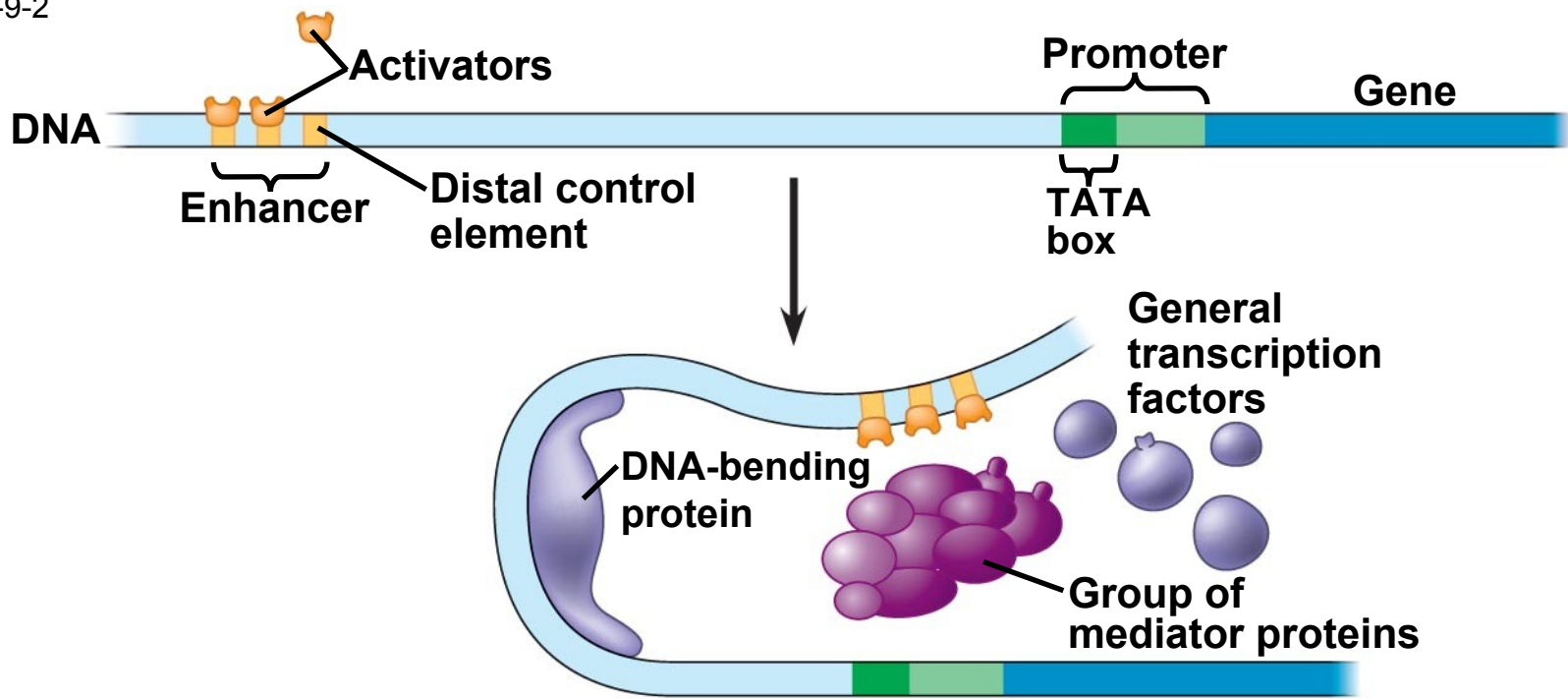
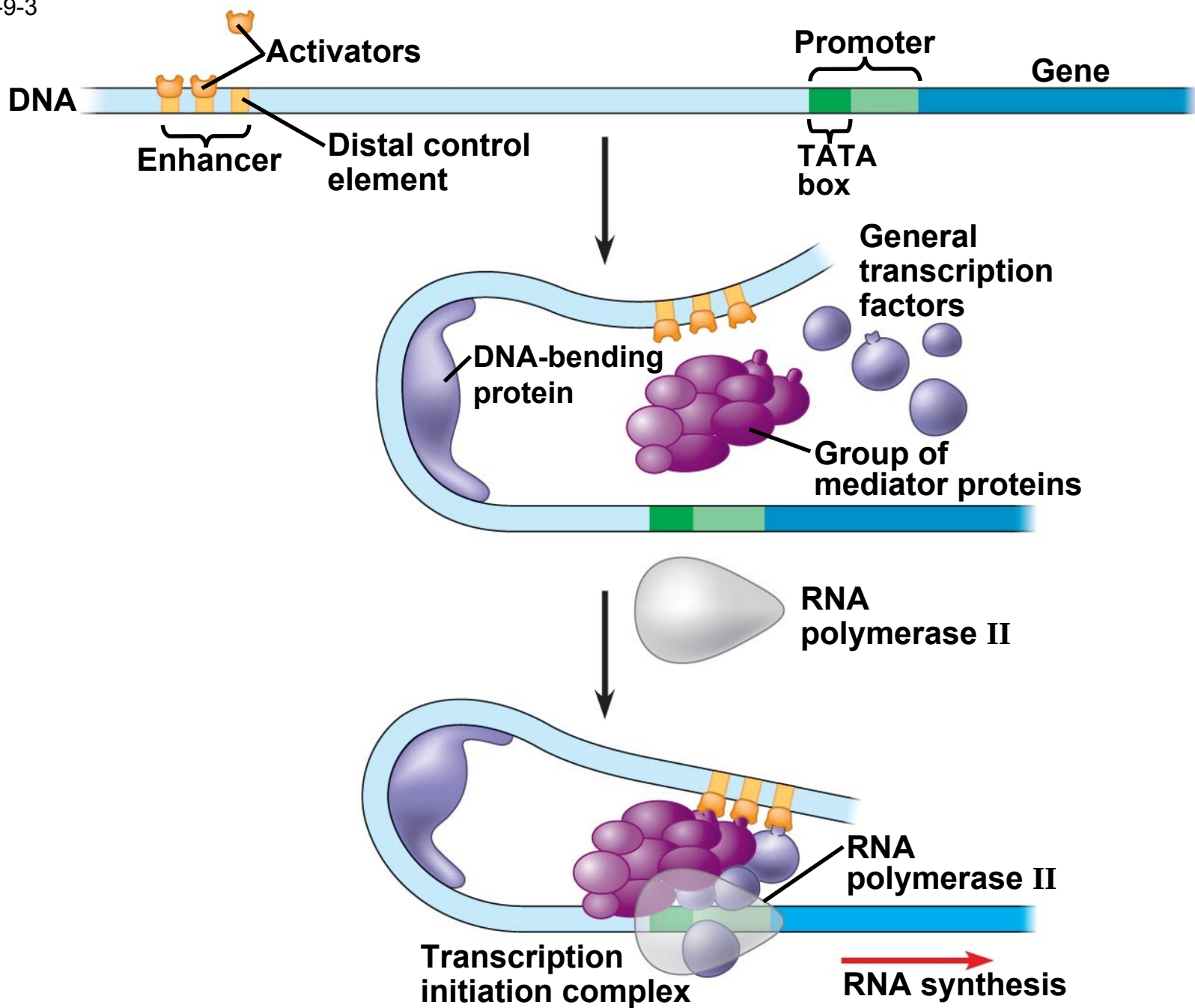
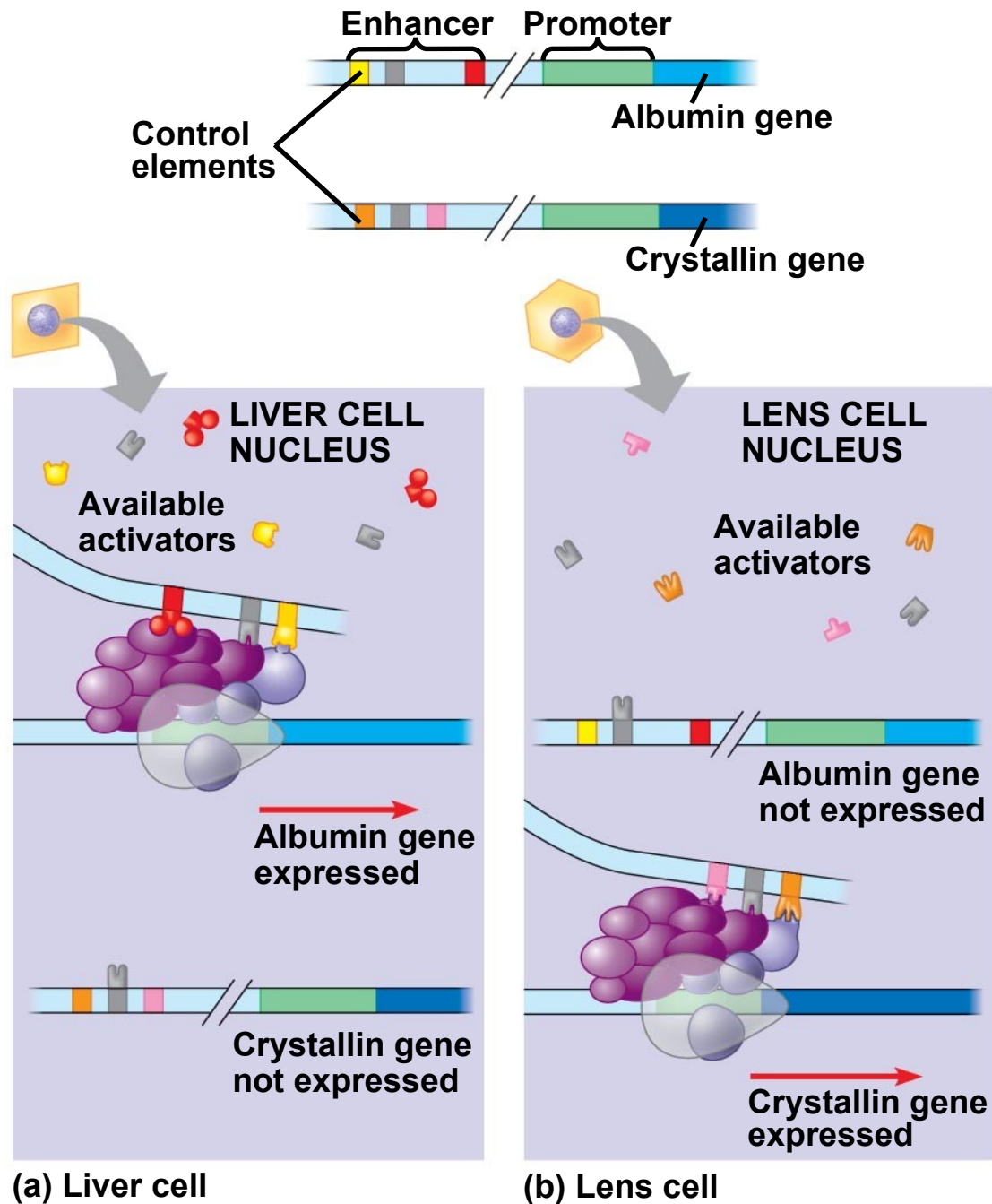


Fig. 18-9-3



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- Some transcription factors function as repressors, inhibiting expression of a particular gene
 - Some activators and repressors act indirectly by influencing chromatin structure to promote or silence transcription

Fig. 18-10



Coordinately Controlled Genes in Eukaryotes

- Unlike the genes of a prokaryotic operon, each of the coordinately controlled eukaryotic genes has a promoter and control elements
- These genes can be scattered over different chromosomes, but each has the same combination of control elements
- Copies of the activators recognize specific control elements and promote simultaneous transcription of the genes

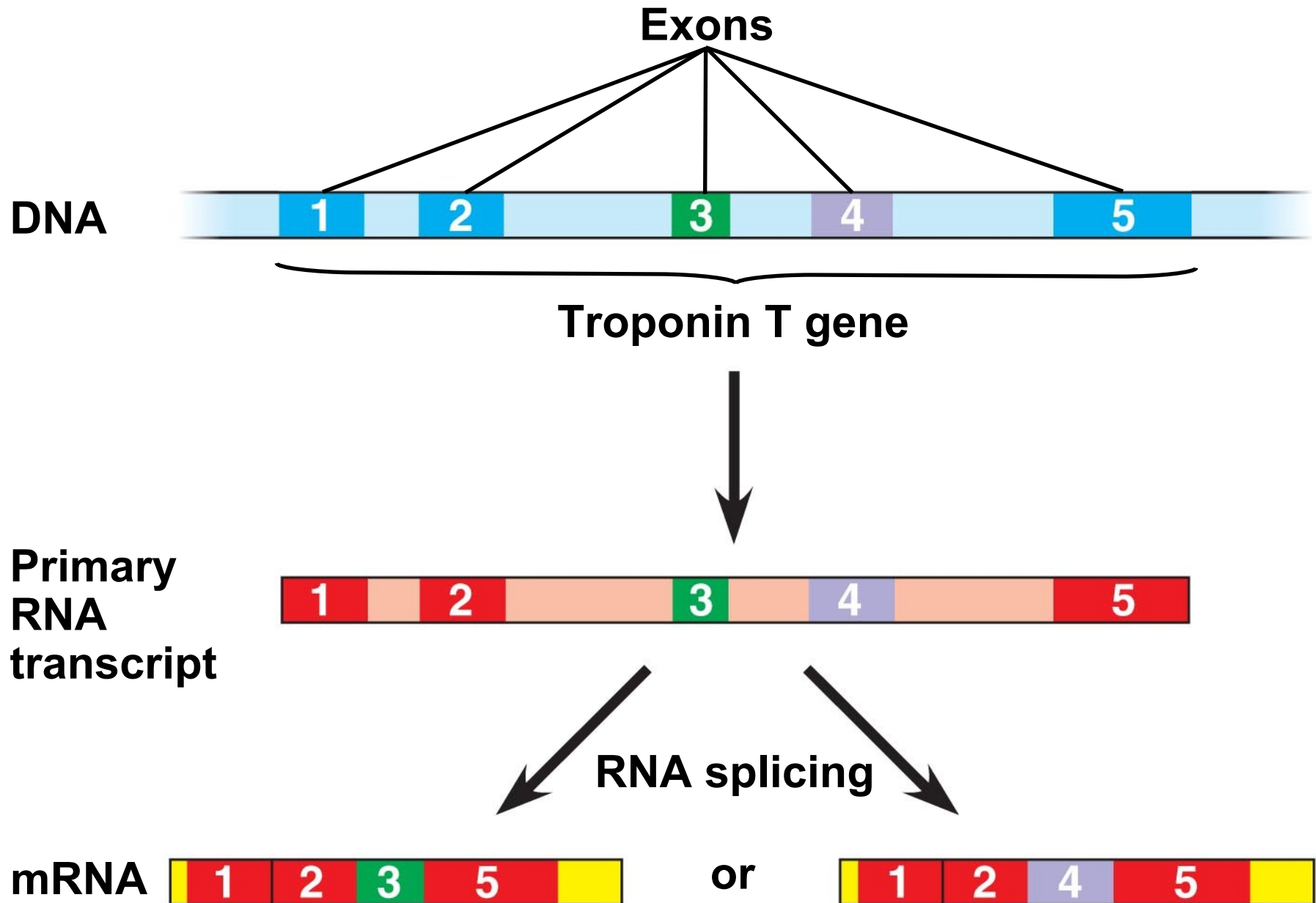
Mechanisms of Post-Transcriptional Regulation

- Transcription alone does not account for gene expression
- Regulatory mechanisms can operate at various stages after transcription
- Such mechanisms allow a cell to fine-tune gene expression rapidly in response to environmental changes

RNA Processing

- In **alternative RNA splicing**, different mRNA molecules are produced from the same primary transcript, depending on which RNA segments are treated as exons and which as introns

Fig. 18-11



mRNA Degradation

- The life span of mRNA molecules in the cytoplasm is a key to determining protein synthesis
- Eukaryotic mRNA is more long lived than prokaryotic mRNA
- The mRNA life span is determined in part by sequences in the leader and trailer regions

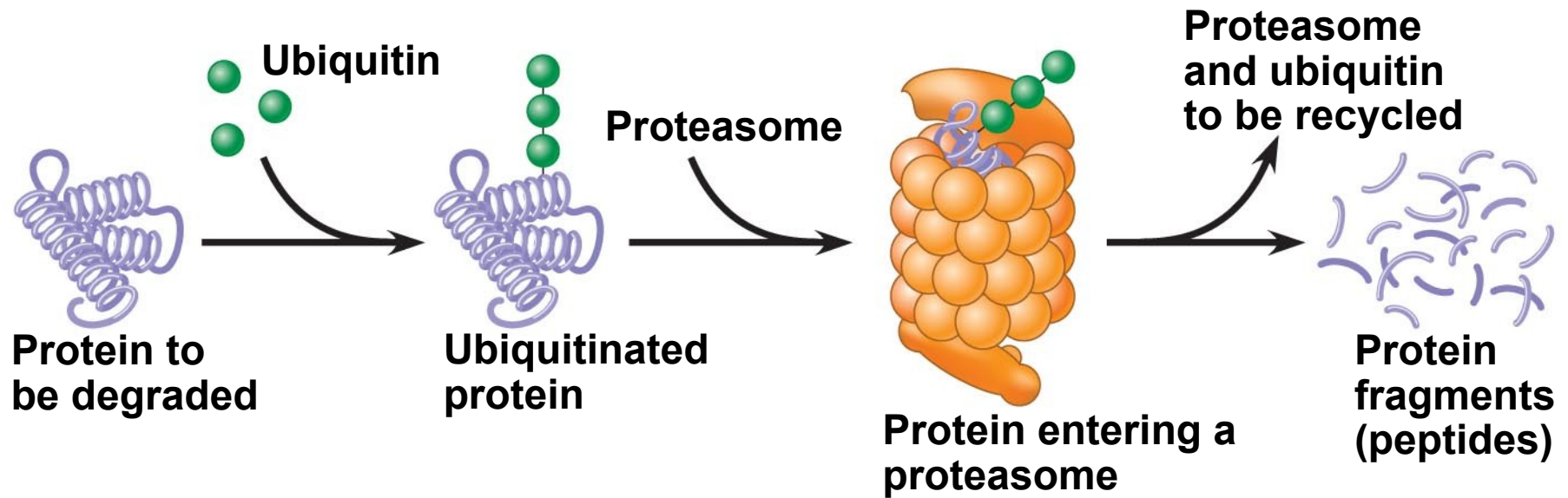
Initiation of Translation

- The initiation of translation of selected mRNAs can be blocked by regulatory proteins that bind to sequences or structures of the mRNA
- Alternatively, translation of all mRNAs in a cell may be regulated simultaneously
- For example, translation initiation factors are simultaneously activated in an egg following fertilization

Protein Processing and Degradation

- After translation, various types of protein processing, including cleavage and the addition of chemical groups, are subject to control
- **Proteasomes** are giant protein complexes that bind protein molecules and degrade them

Fig. 18-12

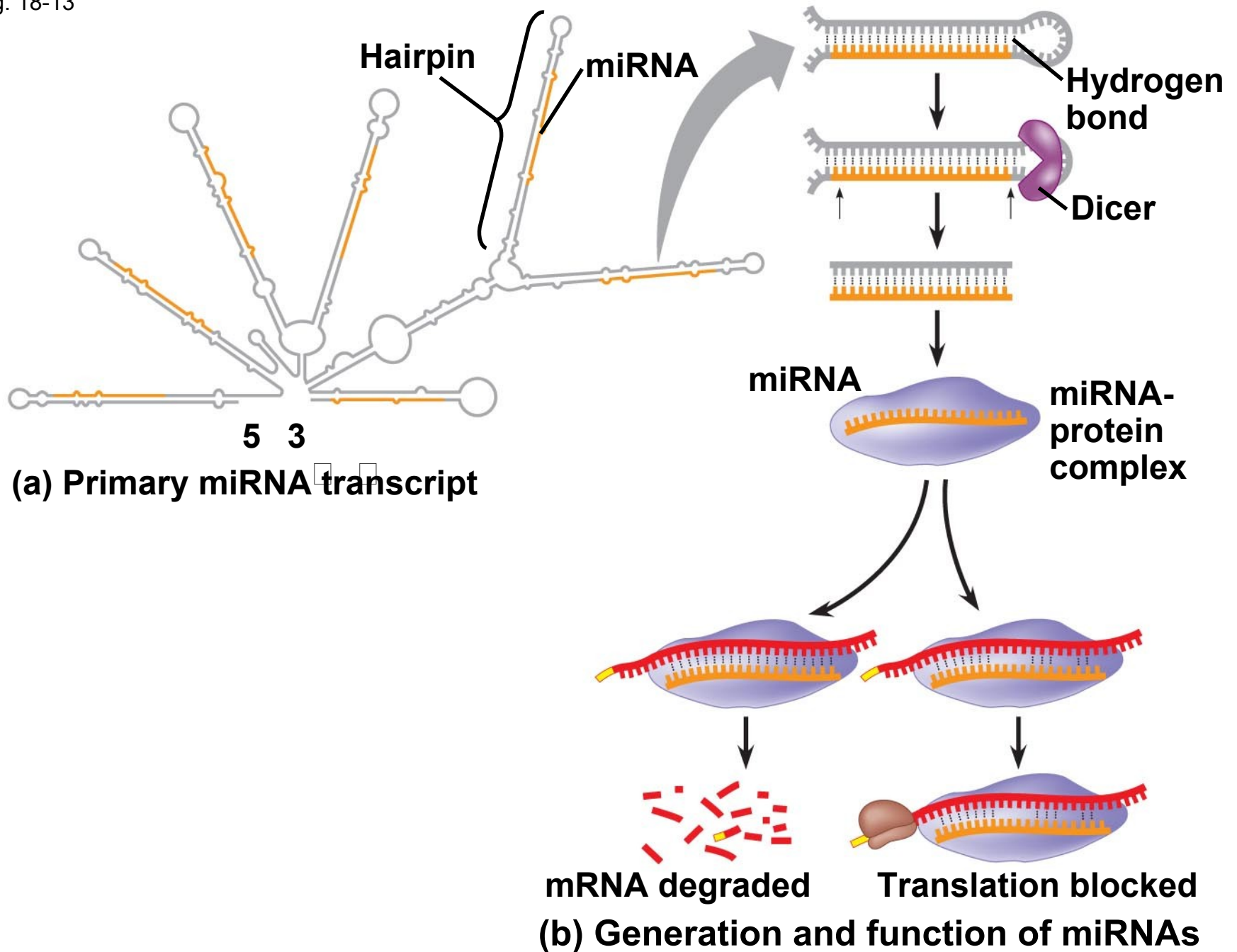


Concept 18.3: Noncoding RNAs play multiple roles in controlling gene expression

- Only a small fraction of DNA codes for proteins, rRNA, and tRNA
- A significant amount of the genome may be transcribed into noncoding RNAs
- Noncoding RNAs regulate gene expression at two points: mRNA translation and chromatin configuration

Effects on mRNAs by MicroRNAs and Small Interfering RNAs

- **MicroRNAs (miRNAs)** are small single-stranded RNA molecules that can bind to mRNA
- These can degrade mRNA or block its translation



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- The phenomenon of inhibition of gene expression by RNA molecules is called **RNA interference (RNAi)**
 - RNAi is caused by **small interfering RNAs (siRNAs)**
 - siRNAs and miRNAs are similar but form from different RNA precursors

Chromatin Remodeling and Silencing of Transcription by Small RNAs

- siRNAs play a role in heterochromatin formation and can block large regions of the chromosome
- Small RNAs may also block transcription of specific genes

Concept 18.4: A program of differential gene expression leads to the different cell types in a multicellular organism

- During embryonic development, a fertilized egg gives rise to many different cell types
- Cell types are organized successively into tissues, organs, organ systems, and the whole organism
- Gene expression orchestrates the developmental programs of animals

A Genetic Program for Embryonic Development

- The transformation from zygote to adult results from cell division, cell differentiation, and morphogenesis



(a) Fertilized eggs of a frog

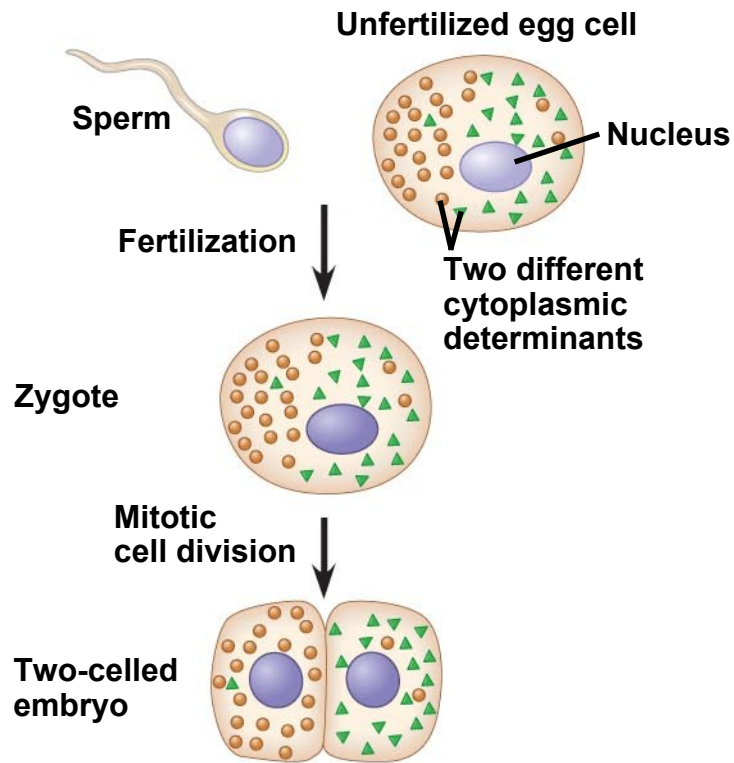


(b) Newly hatched tadpole

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- **Cell differentiation** is the process by which cells become specialized in structure and function
 - The physical processes that give an organism its shape constitute **morphogenesis**
 - Differential gene expression results from genes being regulated differently in each cell type
 - Materials in the egg can set up gene regulation that is carried out as cells divide

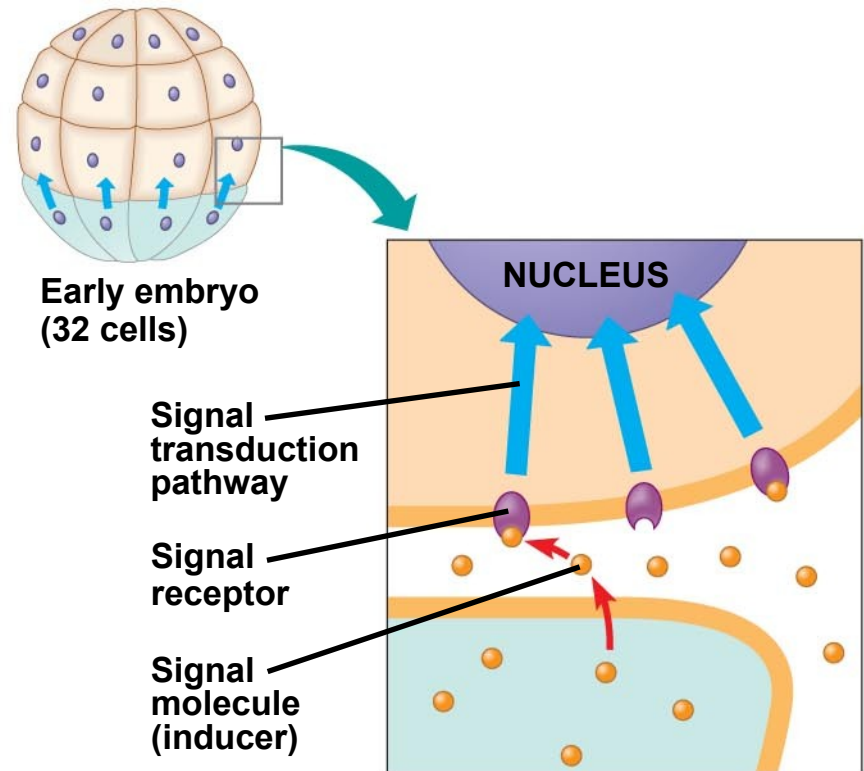
Cytoplasmic Determinants and Inductive Signals

- An egg's cytoplasm contains RNA, proteins, and other substances that are distributed unevenly in the unfertilized egg
- **Cytoplasmic determinants** are maternal substances in the egg that influence early development
- As the zygote divides by mitosis, cells contain different cytoplasmic determinants, which lead to different gene expression

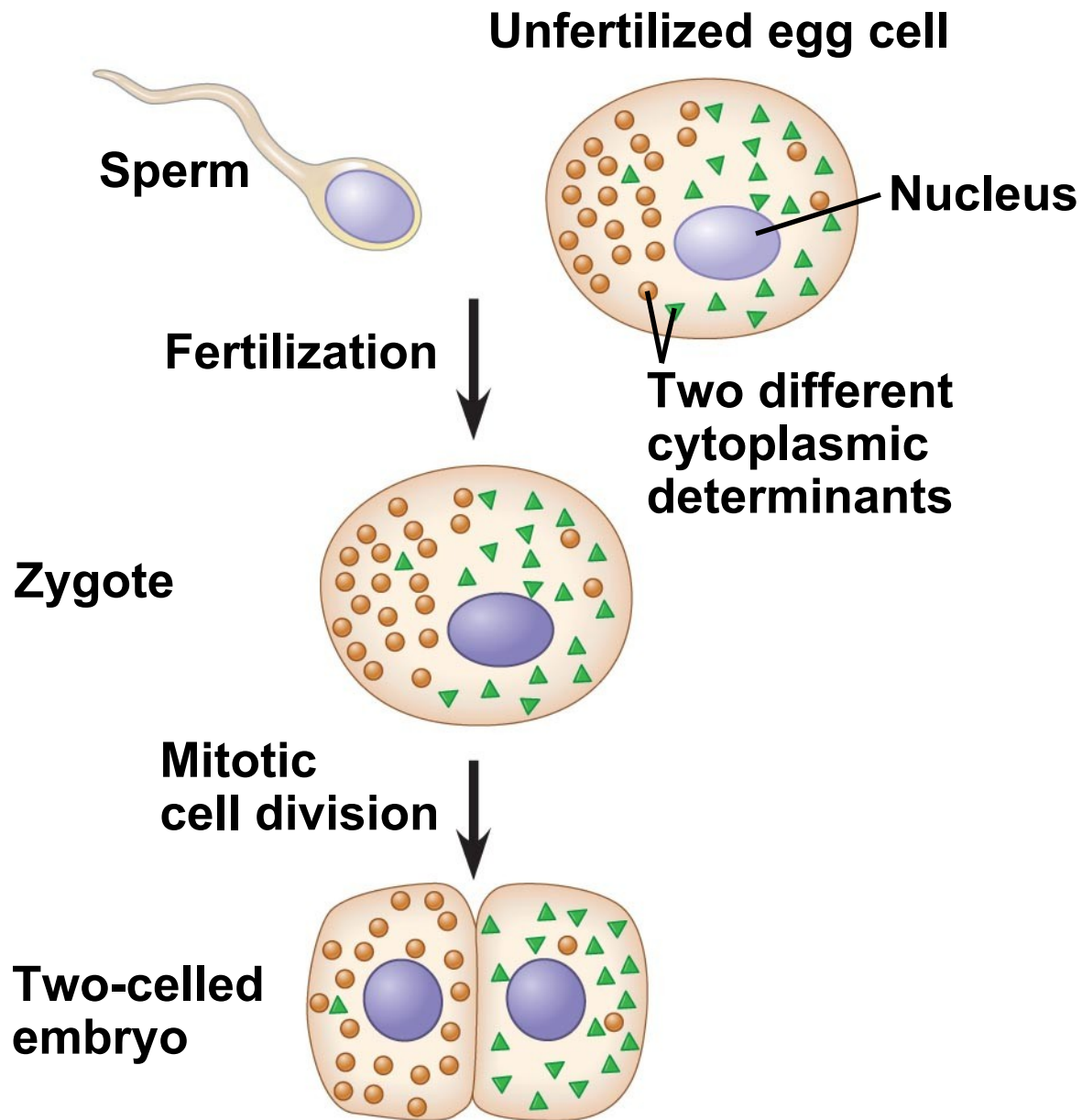


(a) Cytoplasmic determinants in the egg

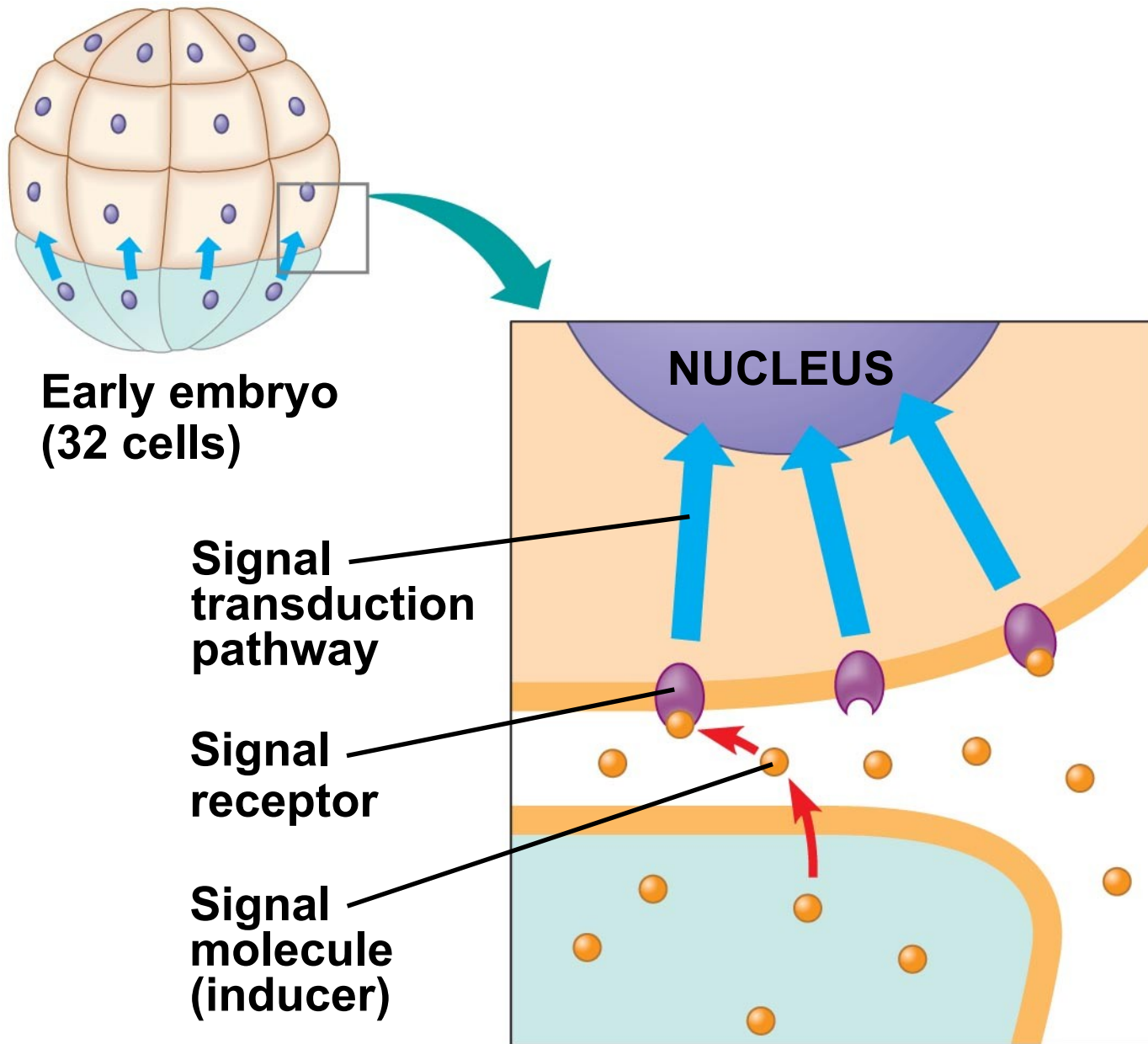
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(b) Induction by nearby cells



(a) Cytoplasmic determinants in the egg



(b) Induction by nearby cells

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- The other important source of developmental information is the environment around the cell, especially signals from nearby embryonic cells
 - In the process called **induction**, signal molecules from embryonic cells cause transcriptional changes in nearby target cells
 - Thus, interactions between cells induce differentiation of specialized cell types

Sequential Regulation of Gene Expression During Cellular Differentiation

- **Determination** commits a cell to its final fate
- Determination precedes differentiation
- Cell differentiation is marked by the production of tissue-specific proteins

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- Myoblasts produce muscle-specific proteins and form skeletal muscle cells
 - *MyoD* is one of several “master regulatory genes” that produce proteins that commit the cell to becoming skeletal muscle
 - The MyoD protein is a transcription factor that binds to enhancers of various target genes

Fig. 18-16-1

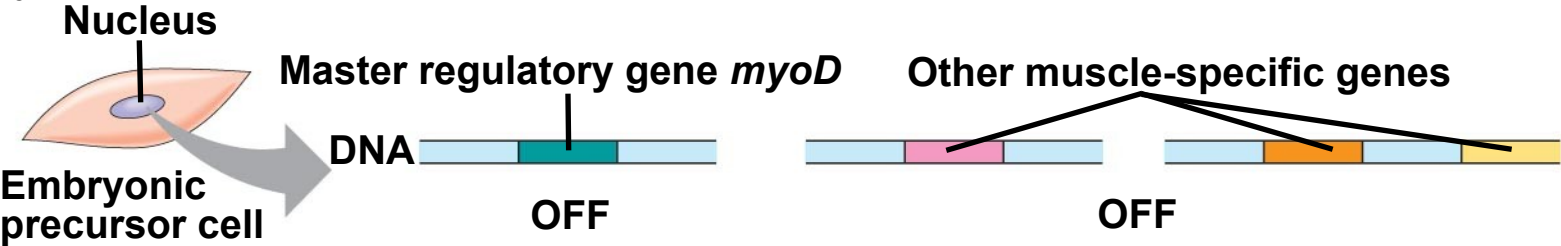


Fig. 18-16-2

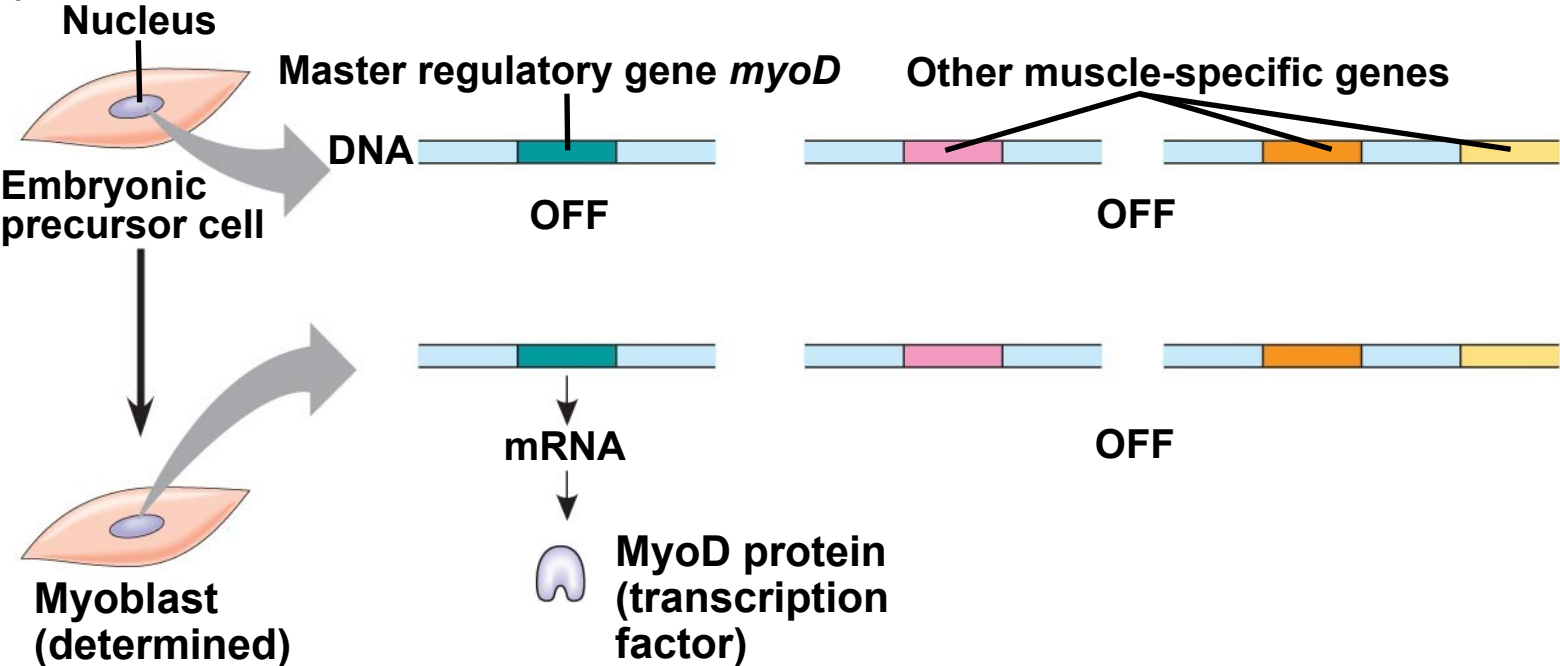
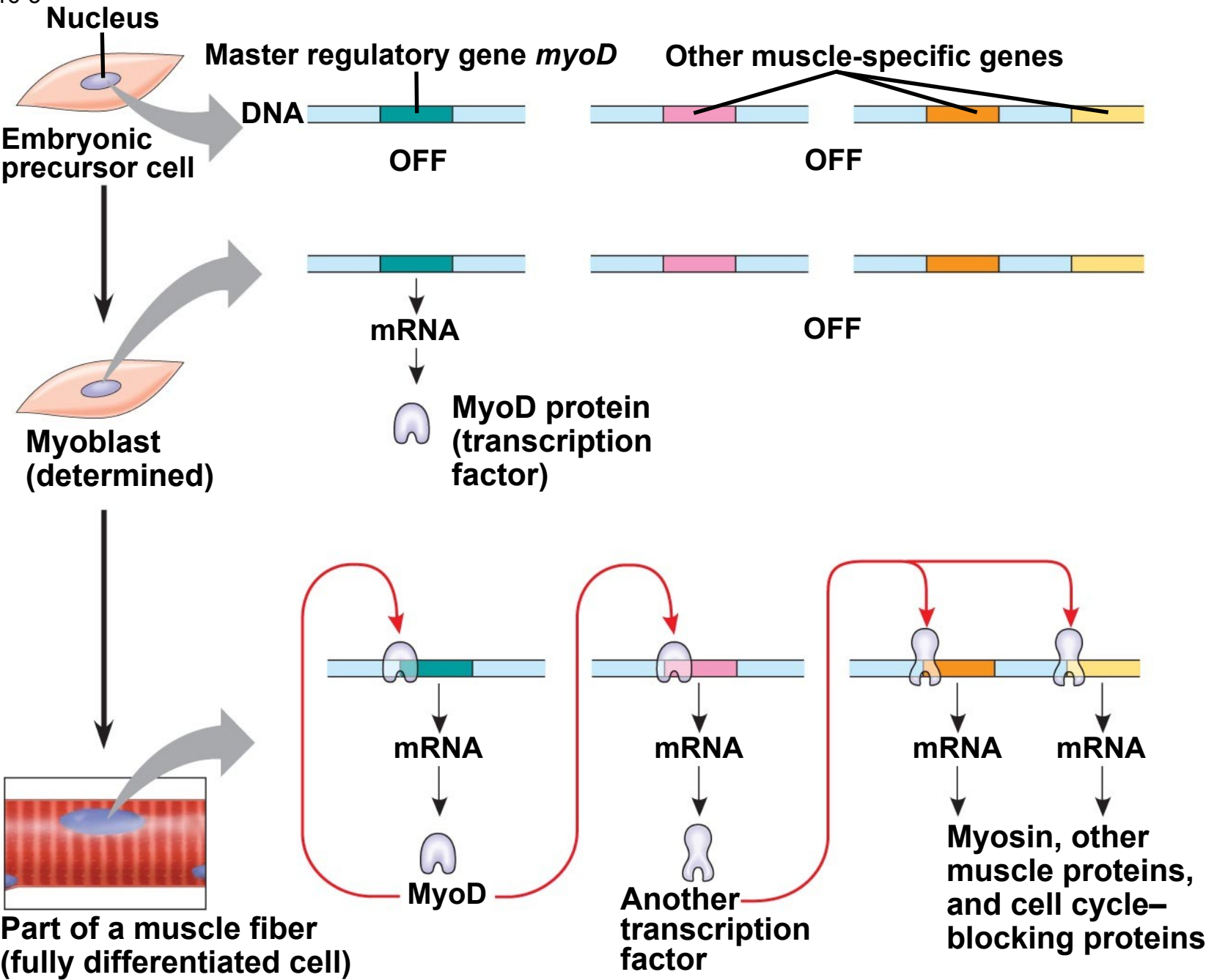


Fig. 18-16-3



Pattern Formation: Setting Up the Body Plan

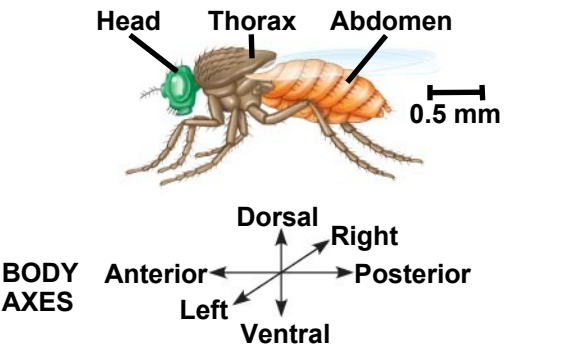
- **Pattern formation** is the development of a spatial organization of tissues and organs
- In animals, pattern formation begins with the establishment of the major axes
- **Positional information**, the molecular cues that control pattern formation, tells a cell its location relative to the body axes and to neighboring cells

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- Pattern formation has been extensively studied in the fruit fly *Drosophila melanogaster*
 - Combining anatomical, genetic, and biochemical approaches, researchers have discovered developmental principles common to many other species, including humans

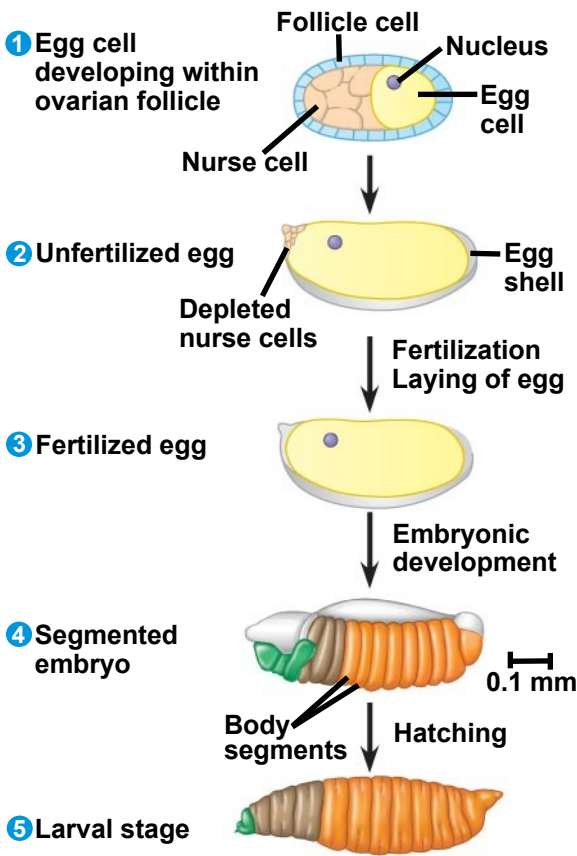
The Life Cycle of Drosophila

- In *Drosophila*, cytoplasmic determinants in the unfertilized egg determine the axes before fertilization
- After fertilization, the embryo develops into a segmented larva with three larval stages

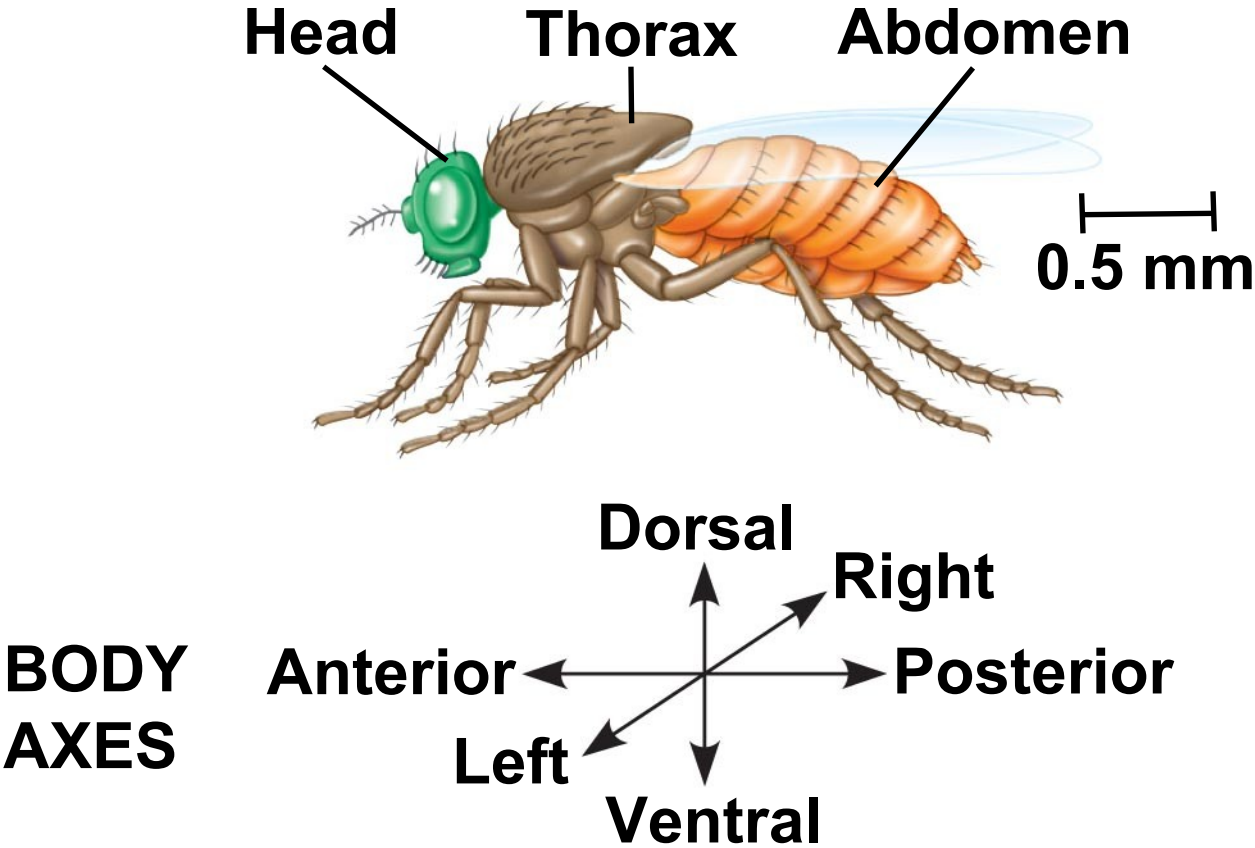
Fig. 18-17



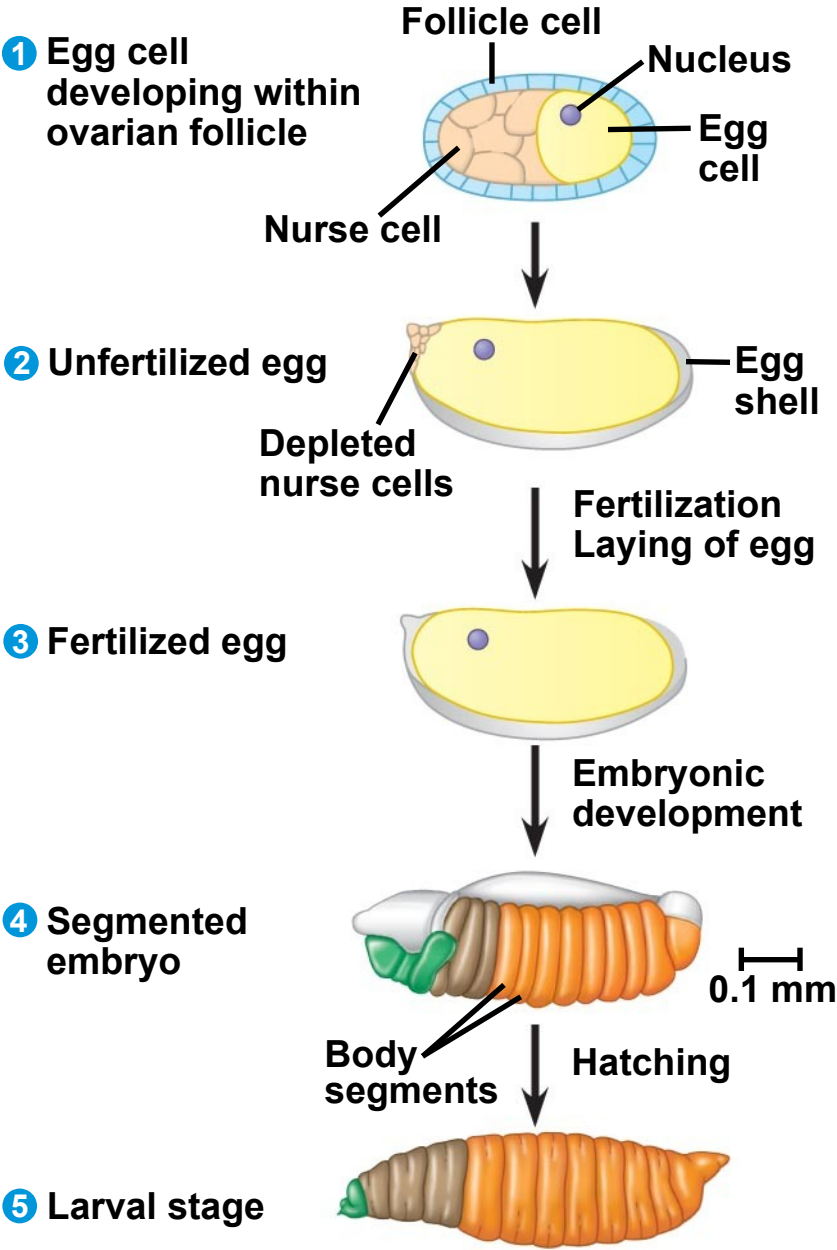
(a) Adult



(b) Development from egg to larva



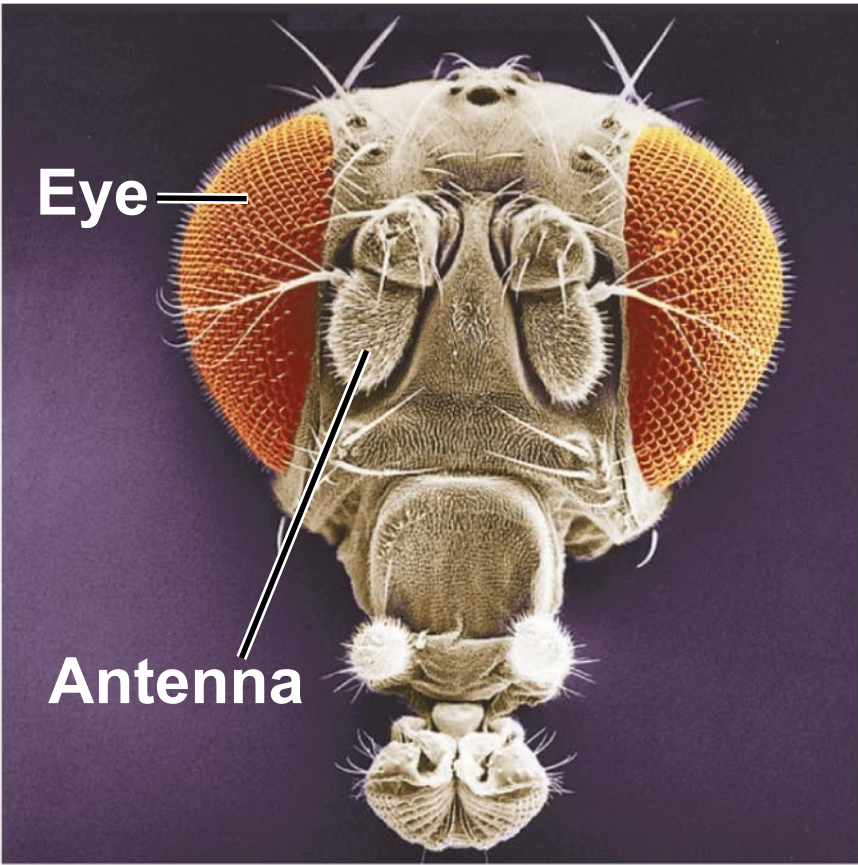
(a) Adult



(b) Development from egg to larva

Genetic Analysis of Early Development: Scientific Inquiry

- Edward B. Lewis, Christiane Nüsslein-Volhard, and Eric Wieschaus won a Nobel 1995 Prize for decoding pattern formation in *Drosophila*
- Lewis demonstrated that genes direct the developmental process

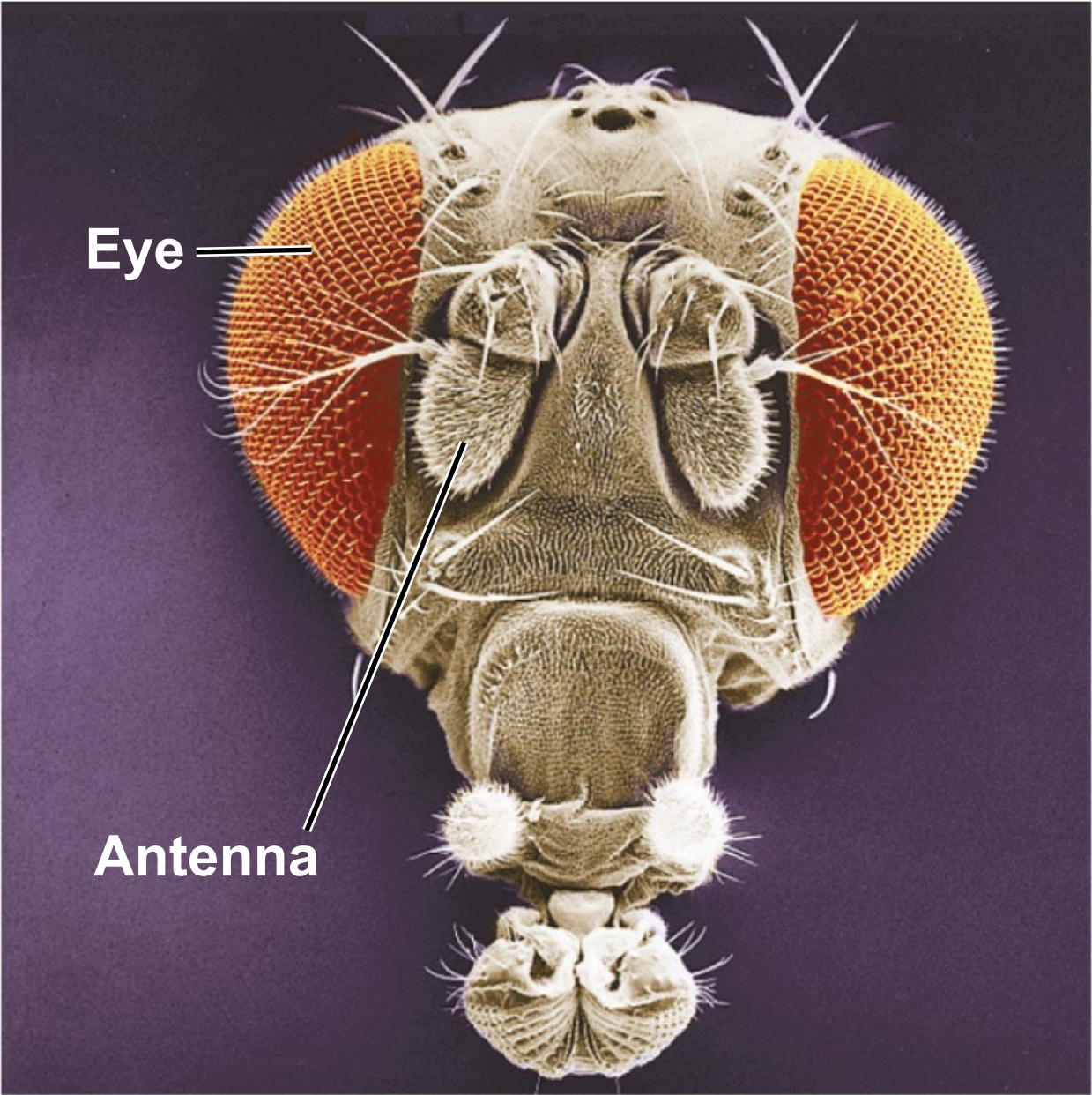


Wild type



Mutant

Fig. 18-18a

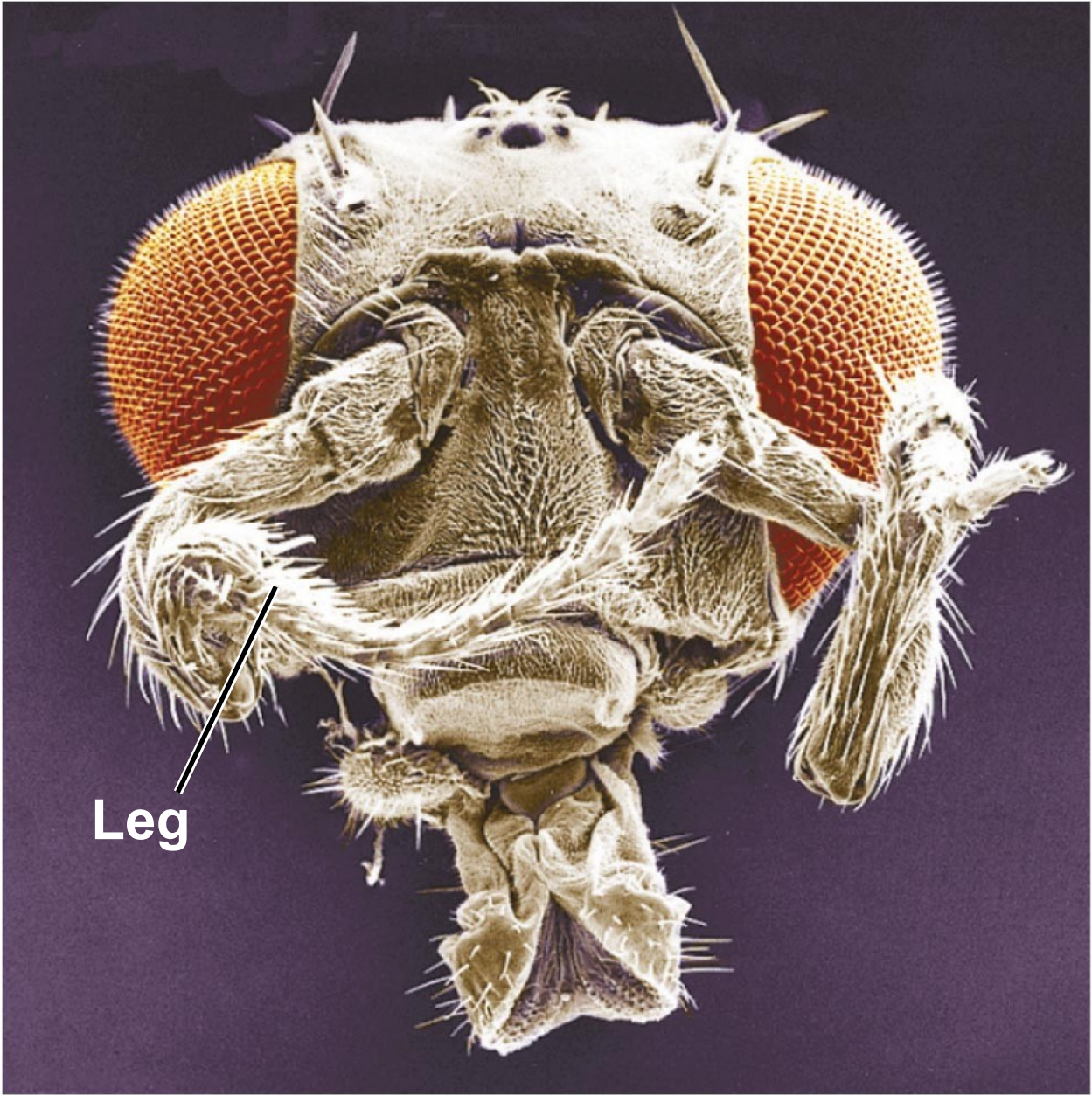


Eye

Antenna

Wild type

Fig. 18-18b



Mutant

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- Nüsslein-Volhard and Wieschaus studied segment formation
 - They created mutants, conducted breeding experiments, and looked for corresponding genes
 - Breeding experiments were complicated by **embryonic lethals**, embryos with lethal mutations
 - They found 120 genes essential for normal segmentation

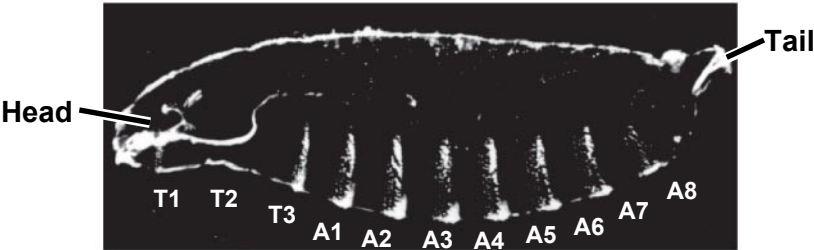
Axis Establishment

- **Maternal effect genes** encode for cytoplasmic determinants that initially establish the axes of the body of *Drosophila*
- These maternal effect genes are also called **egg-polarity genes** because they control orientation of the egg and consequently the fly

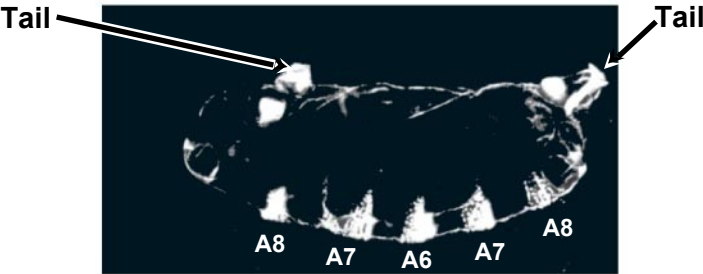
Bicoid: A Morphogen Determining Head Structures

- One maternal effect gene, the ***bicoid*** gene, affects the front half of the body
- An embryo whose mother has a mutant *bicoid* gene lacks the front half of its body and has duplicate posterior structures at both ends

EXPERIMENT

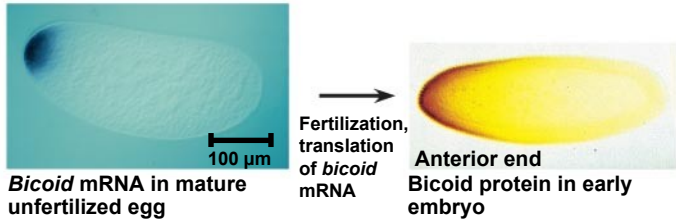


Wild-type larva



Mutant larva
(*bicoid*)

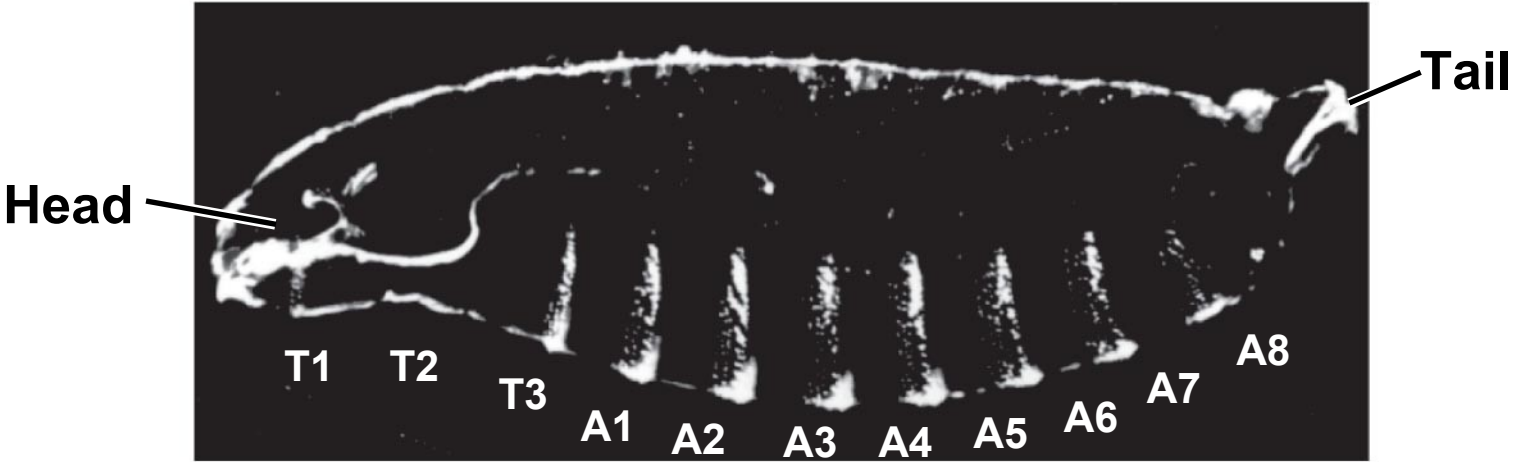
RESULTS



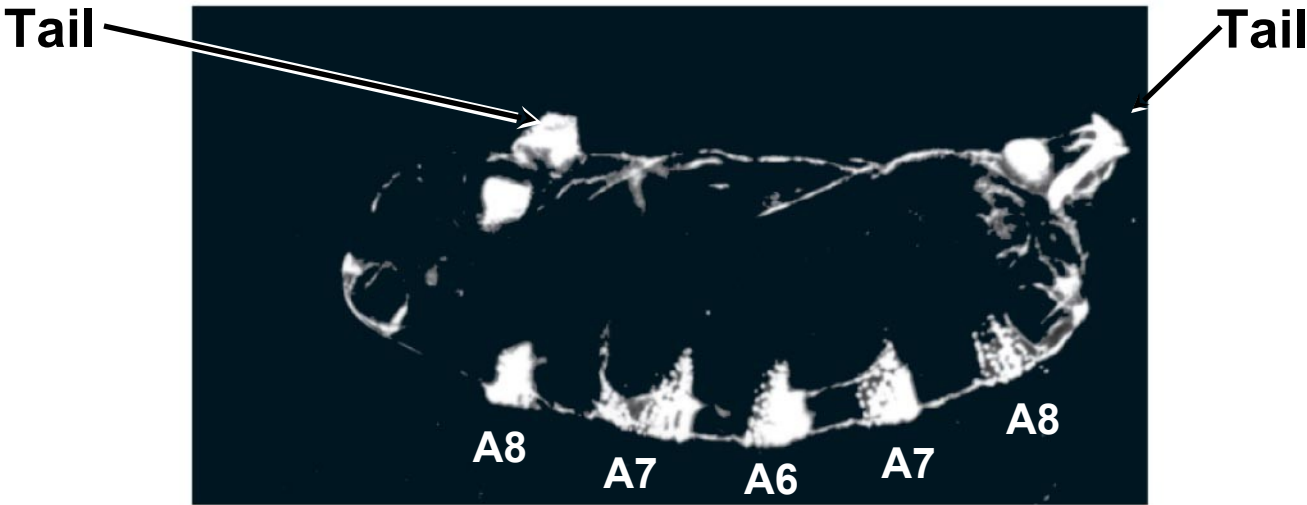
CONCLUSION



EXPERIMENT

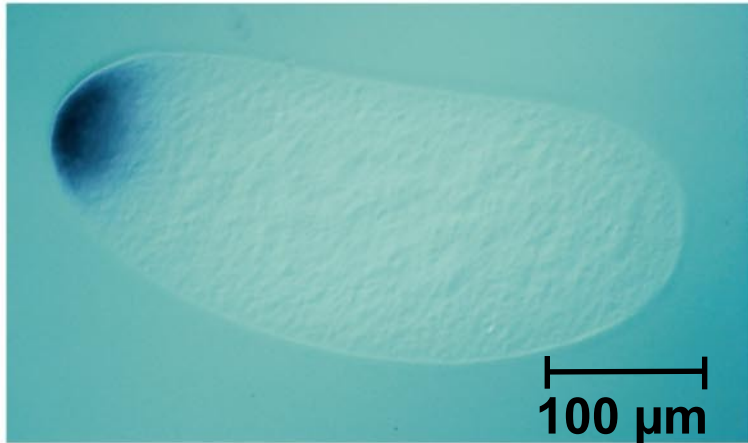


Wild-type larva



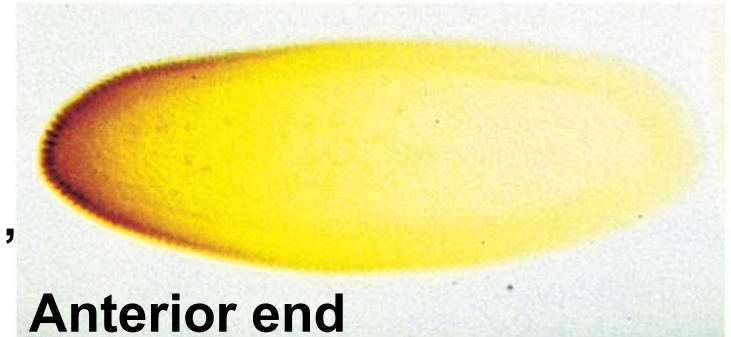
Mutant larva (*bicoid*)

RESULTS



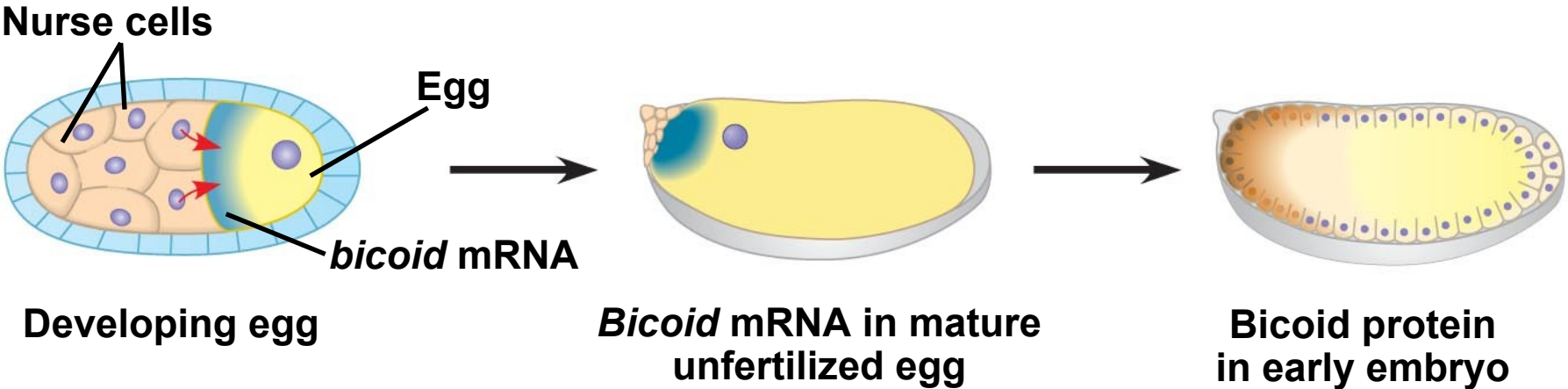
***Bicoid* mRNA in mature unfertilized egg**

→
**Fertilization,
translation
of *bicoid*
mRNA**



**Anterior end
Bicoid protein in early embryo**

CONCLUSION



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- This phenotype suggests that the product of the mother's *bicoid* gene is concentrated at the future anterior end
 - This hypothesis is an example of the gradient hypothesis, in which gradients of substances called **morphogens** establish an embryo's axes and other features

-
- The *bicoid* research is important for three reasons:
 - It identified a specific protein required for some early steps in pattern formation
 - It increased understanding of the mother's role in embryo development
 - It demonstrated a key developmental principle that a gradient of molecules can determine polarity and position in the embryo

Concept 18.5: Cancer results from genetic changes that affect cell cycle control

- The gene regulation systems that go wrong during cancer are the very same systems involved in embryonic development

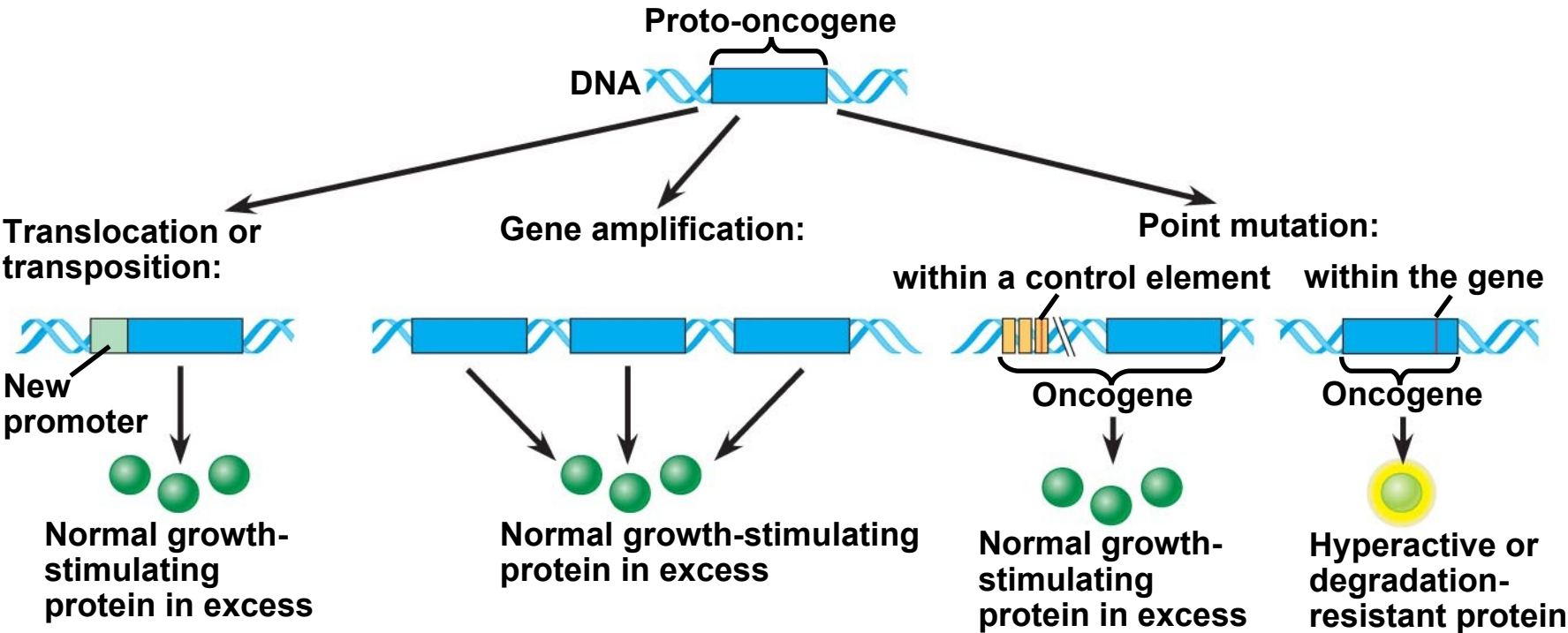
Types of Genes Associated with Cancer

- Cancer can be caused by mutations to genes that regulate cell growth and division
- Tumor viruses can cause cancer in animals including humans

Oncogenes and Proto-Oncogenes

- **Oncogenes** are cancer-causing genes
- **Proto-oncogenes** are the corresponding normal cellular genes that are responsible for normal cell growth and division
- Conversion of a proto-oncogene to an oncogene can lead to abnormal stimulation of the cell cycle

Fig. 18-20



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- Proto-oncogenes can be converted to oncogenes by
 - Movement of DNA within the genome: if it ends up near an active promoter, transcription may increase
 - Amplification of a proto-oncogene: increases the number of copies of the gene
 - Point mutations in the proto-oncogene or its control elements: causes an increase in gene expression

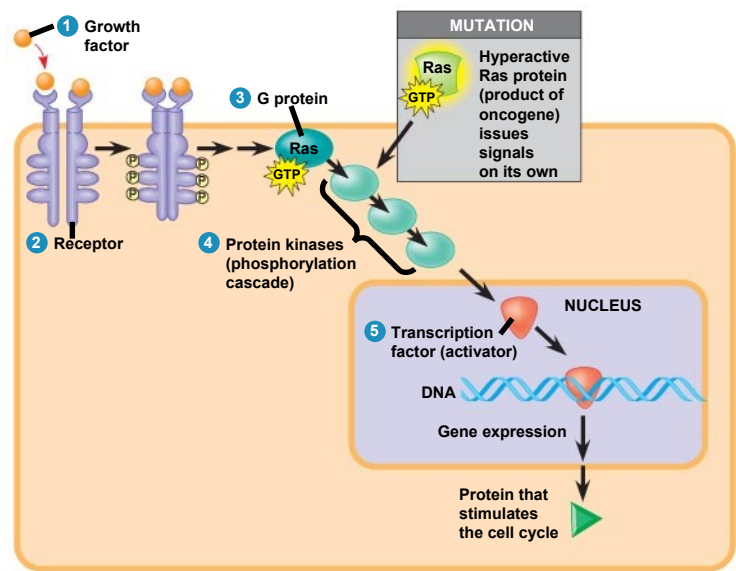
Tumor-Suppressor Genes

- **Tumor-suppressor genes** help prevent uncontrolled cell growth
- Mutations that decrease protein products of tumor-suppressor genes may contribute to cancer onset
- Tumor-suppressor proteins
 - Repair damaged DNA
 - Control cell adhesion
 - Inhibit the cell cycle in the cell-signaling pathway

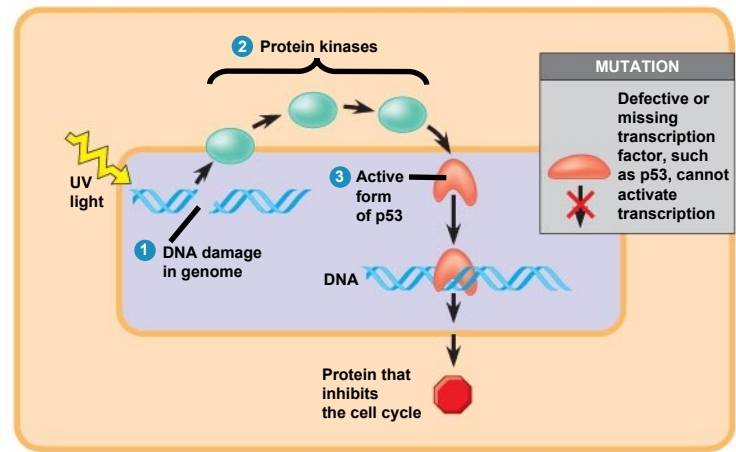
Interference with Normal Cell-Signaling Pathways

- Mutations in the *ras* proto-oncogene and *p53* tumor-suppressor gene are common in human cancers
- Mutations in the ***ras* gene** can lead to production of a hyperactive Ras protein and increased cell division

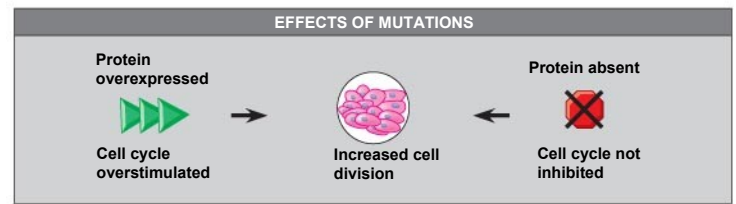
Fig. 18-21



(a) Cell cycle–stimulating pathway

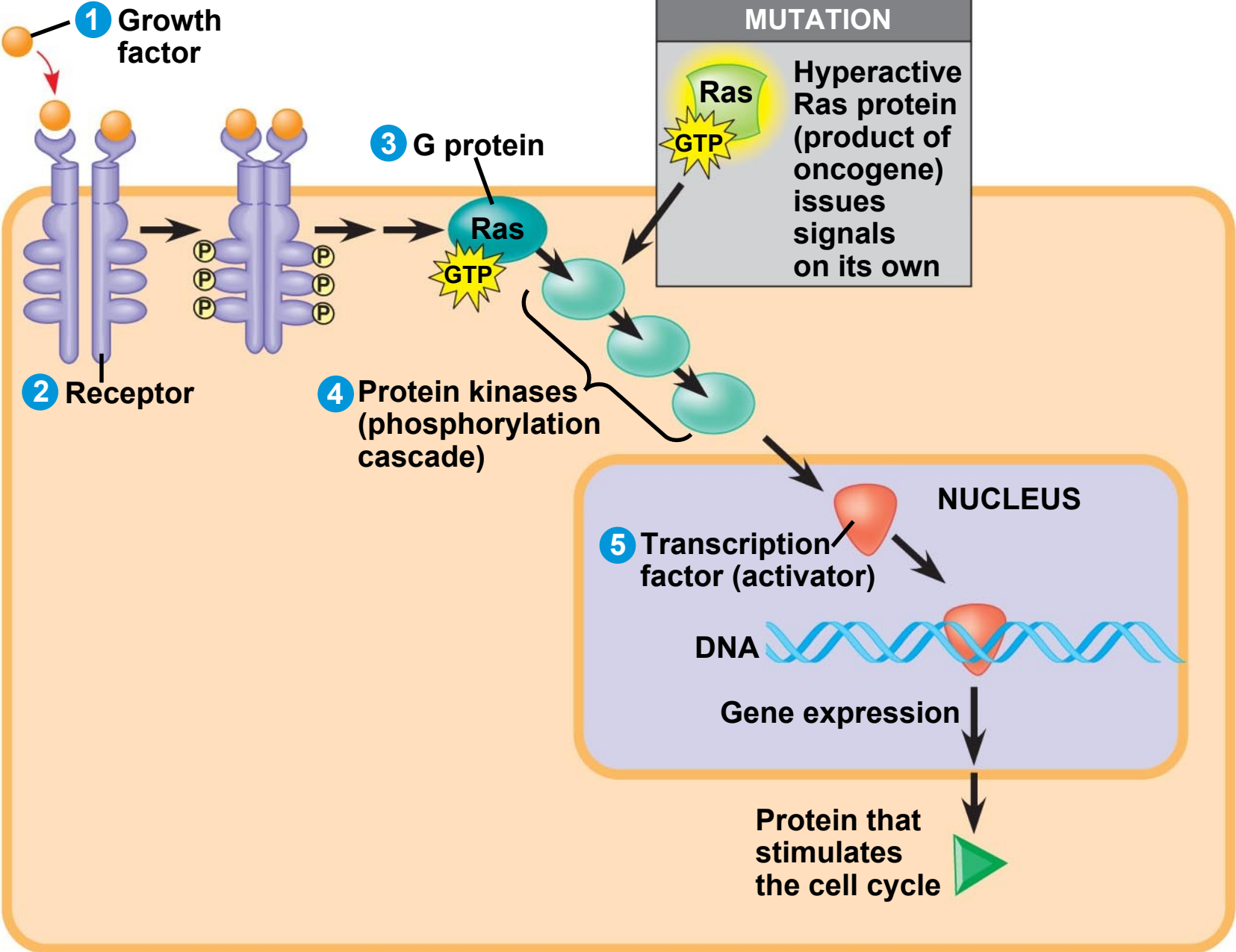


(b) Cell cycle–inhibiting pathway



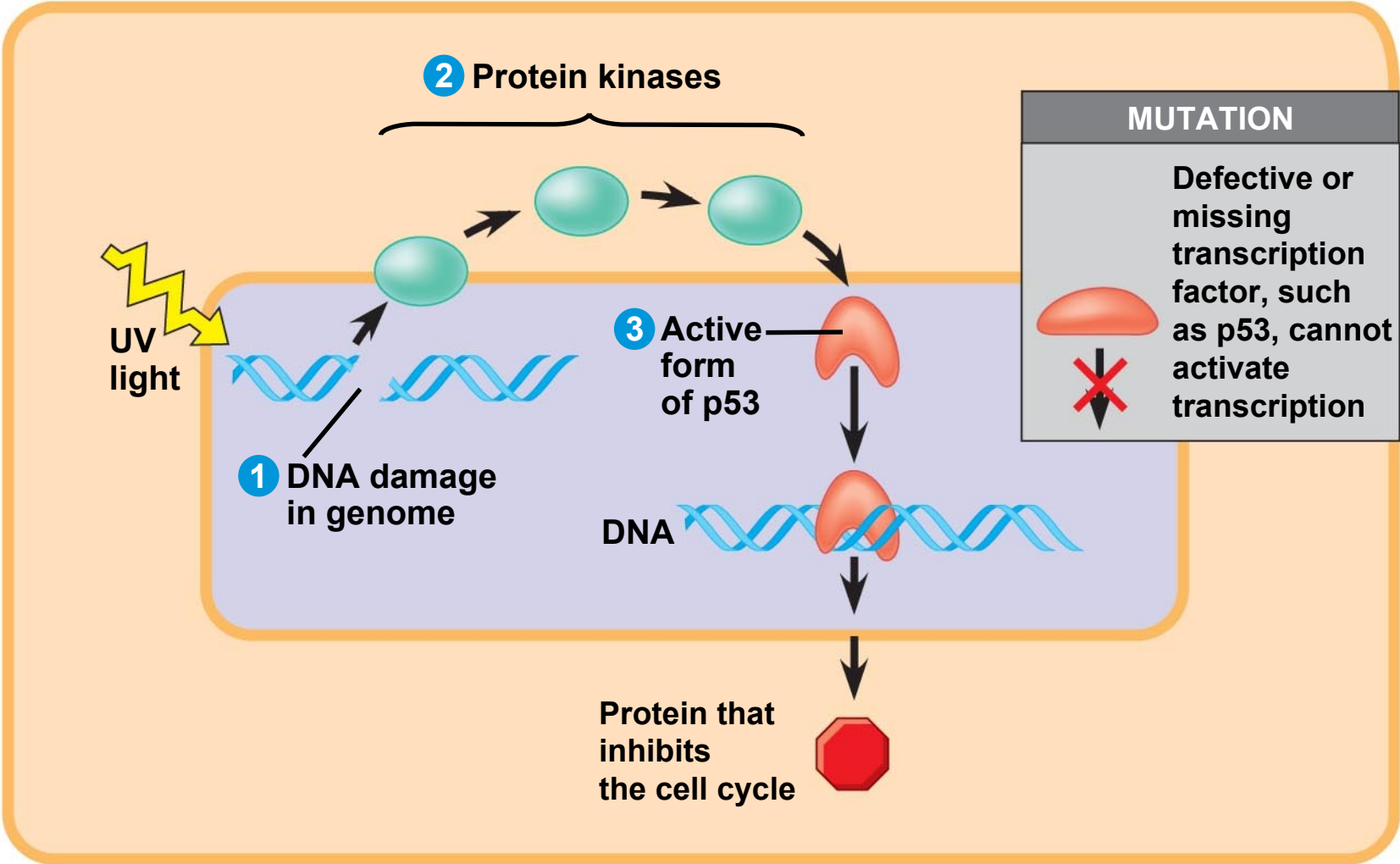
(c) Effects of mutations

Fig. 18-21a

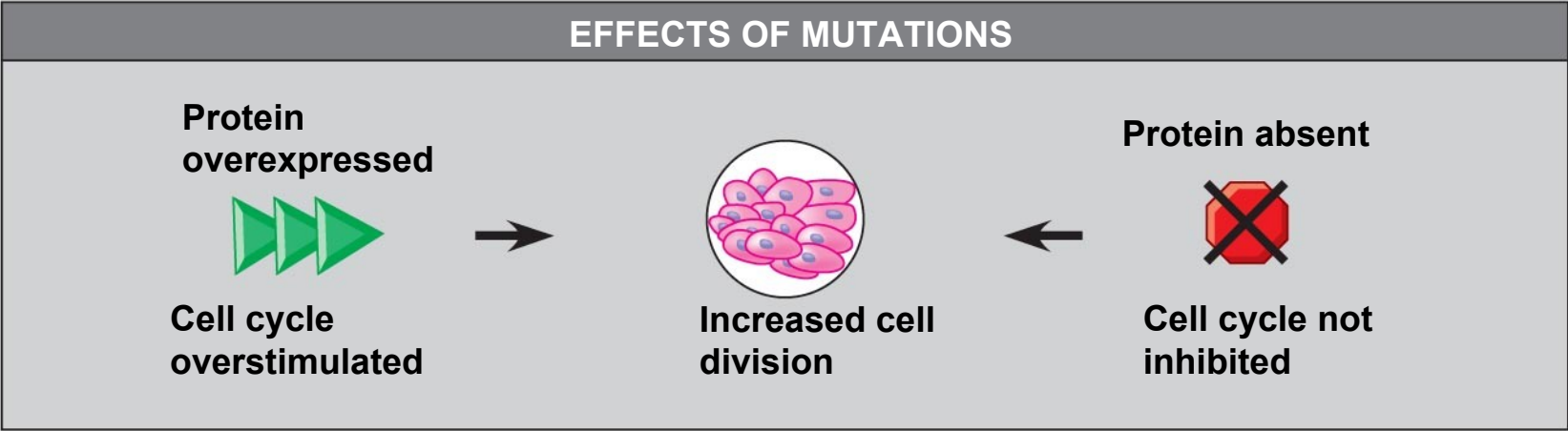


(a) Cell cycle–stimulating pathway

Fig. 18-21b



(b) Cell cycle–inhibiting pathway



(c) Effects of mutations

-
- Suppression of the cell cycle can be important in the case of damage to a cell's DNA; *p53* prevents a cell from passing on mutations due to DNA damage
 - Mutations in the ***p53* gene** prevent suppression of the cell cycle

The Multistep Model of Cancer Development

- Multiple mutations are generally needed for full-fledged cancer; thus the incidence increases with age
- At the DNA level, a cancerous cell is usually characterized by at least one active oncogene and the mutation of several tumor-suppressor genes

Fig. 18-22

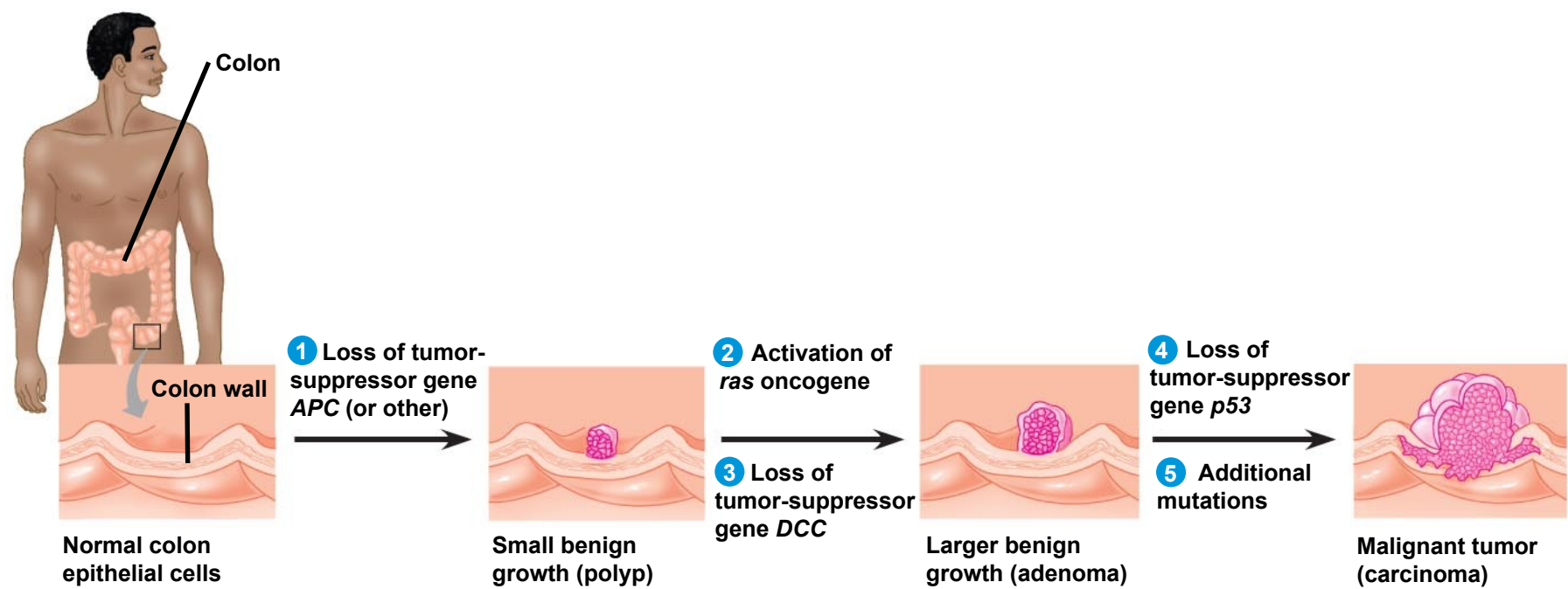
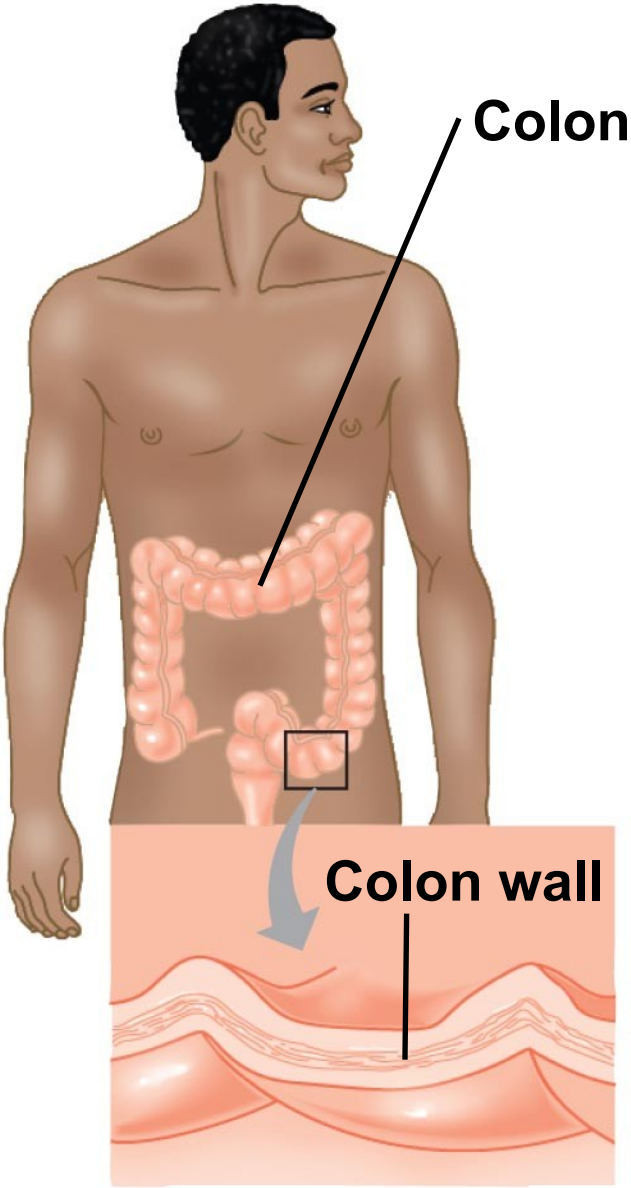
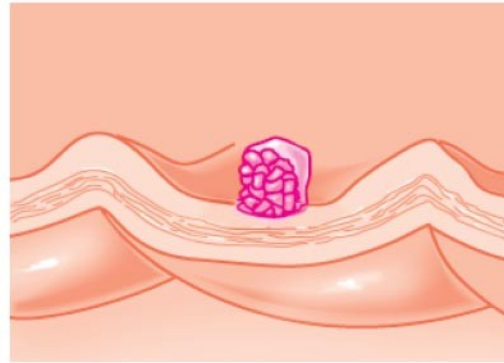


Fig. 18-22a



**Normal colon
epithelial cells**

1 Loss of tumor-suppressor gene *APC* (or other)

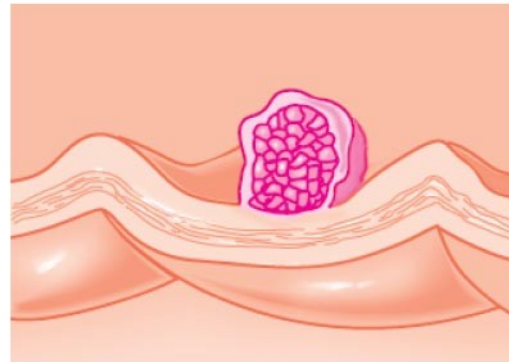


**Small benign
growth (polyp)**

2 Activation of
ras oncogene



3 Loss of
tumor-suppressor
gene *DCC*

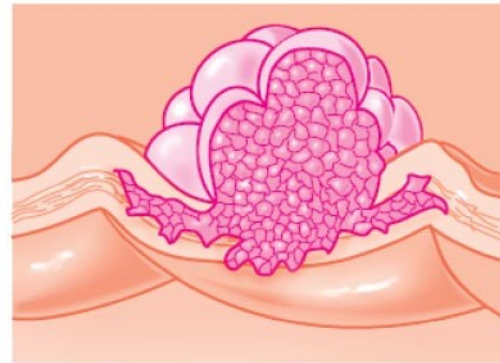


**Larger benign
growth (adenoma)**

4 Loss of
tumor-suppressor
gene *p53*



5 Additional
mutations

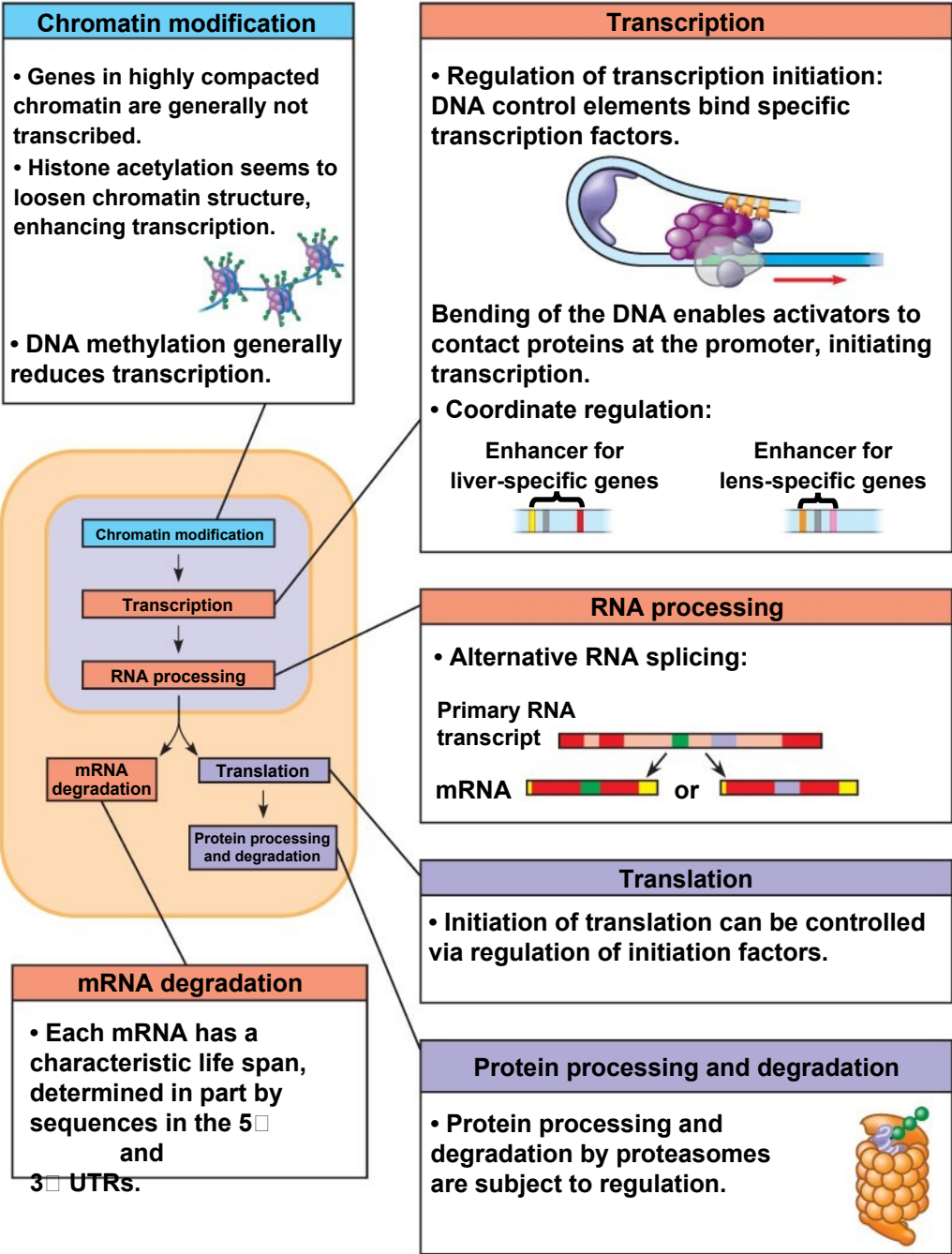


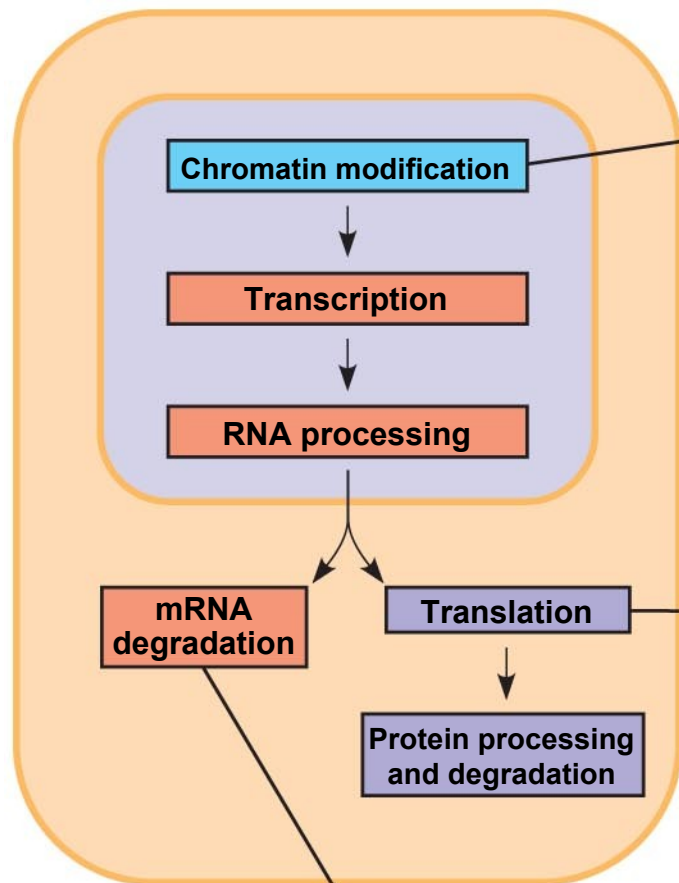
**Malignant tumor
(carcinoma)**

Inherited Predisposition and Other Factors Contributing to Cancer

- Individuals can inherit oncogenes or mutant alleles of tumor-suppressor genes
- Inherited mutations in the tumor-suppressor gene *adenomatous polyposis coli* are common in individuals with colorectal cancer
- Mutations in the BRCA1 or BRCA2 gene are found in at least half of inherited breast cancers

Fig. 18-UN4



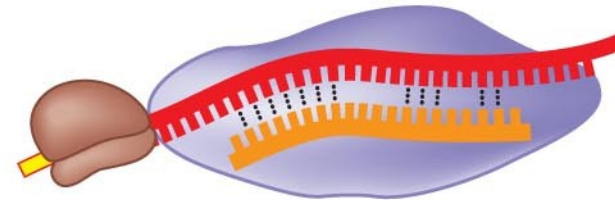


Chromatin modification

- Small RNAs can promote the formation of heterochromatin in certain regions, blocking transcription.

Translation

- miRNA or siRNA can block the translation of specific mRNAs.



mRNA degradation

- miRNA or siRNA can target specific mRNAs for destruction.

You should now be able to:

1. Explain the concept of an operon and the function of the operator, repressor, and corepressor
2. Explain the adaptive advantage of grouping bacterial genes into an operon
3. Explain how repressible and inducible operons differ and how those differences reflect differences in the pathways they control

-
4. Explain how DNA methylation and histone acetylation affect chromatin structure and the regulation of transcription
 5. Define control elements and explain how they influence transcription
 6. Explain the role of promoters, enhancers, activators, and repressors in transcription control

-
7. Explain how eukaryotic genes can be coordinately expressed
 8. Describe the roles played by small RNAs on gene expression
 9. Explain why determination precedes differentiation
 10. Describe two sources of information that instruct a cell to express genes at the appropriate time

-
11. Explain how maternal effect genes affect polarity and development in *Drosophila* embryos
 12. Explain how mutations in tumor-suppressor genes can contribute to cancer
 13. Describe the effects of mutations to the *p53* and *ras* genes